

TOPICAL REVIEW

Visual Deepfake Detection: Review of Techniques, Tools, Limitations, and Future Prospects

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ABSTRACT In recent years, rapid advancements in deepfakes (incorporating Artificial Intelligence (AI), machine, and deep learning) have updated tools and techniques for manipulating multimedia. Though technology has primarily been utilized for beneficial purposes, such as education and entertainment, it is also used for malicious or unethical tasks to spread disinformation or ruin someone's dignity, even if it encompasses harassing and blackmailing victims. Deepfakes refer to high-quality and realistic multimedia-manipulated content that has been digitally modified or synthetically generated. We conducted a systematic literature review of deepfake detection to offer an updated overview of existing research work that initially describes the widely accessible deepfake generation tools, classifications, and detection process. We highlighted recent techniques in visual deepfake detection based on the feature representations, grouped into four domains: spatial, temporal, frequency, and spatio-temporal, including their key features and limitations by providing details of existing datasets, together with the potentials of deepfake and its future directions. This study tried to add an updated repository of technological change in deepfake, which could help researchers to develop robust deepfake models.

INDEX TERMS Deepfake detection, machine learning, deep learning, deepfake applications, deepfake datasets, deepfake generation tools.

I. INTRODUCTION

In 2017, an anonymous Reddit user applied deep learning algorithms to replace a single individual's face in pornographic movies with another, producing fake videos that were remarkably realistic [1]. After that, in January 2018, several websites, with the support of some private sponsors, introduced a deepfake creation service. A month later, these services were prohibited by several websites, including Twitter, Pornhub, and Gfycat [2]. However, the research on deepfake content is very quick in light of the dangers and possible hazards associated with privacy concerns. Images and videos were manipulated in the form of fake content using

AI, which had been announced as a trendy field in the past years [3].

The term "deepfake" is referred as an AI based technology synthesizing the media [4]. The recent generation of fake material, such as counterfeit images and videos, with the help of artificial approaches, has become more common. Deepfake combines two literals, Deep and fake, related to fake material generated by Deep Neural Network (DNN)s, [5]. The audio/video generated data via the deepfake technique is genuine and realistic (as we can see in Figure 1) and not easily recognizable by the human being. In deepfake content, mostly facial features are swapped from one person with the targeted person in video or images and fabricated material to provide the false impression that the target has said mimicked that someone else has [6]. Deepfakes are predicted to take current sources of misinformation and

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disinformation to new heights, making them vulnerable to manipulation by foreign governments, trolls, bots, conspiracy theorists, and hyperpartisan media known as fake news 2.0 [7].

Truth-finding in the digital space has thus grown more crucial. Dealing with deepfakes is far more challenging because they are typically used for malicious purposes and can now be simply made by almost anyone using available deepfake tools. To date, many strategies for detecting deepfakes have been developed. Since deep learning is the foundation of the majority of them, a conflict has grown between the beneficial and harmful applications of deep learning techniques [8].

The United States (US) Defense Advanced Research Projects Agency (DARPA) launched the MediFor research or media forensics program to speed up the advancement of fake digital visual media detection techniques in response to risks posed by deepfakes technologies [9]. The deepfake detection was recently introduced by Facebook Inc., Microsoft Corp., and the partnership on AI coalition to stimulate more research and development in the identification and prevention of deepfakes being utilized by deceptive viewers [10]. By using deep learning technology, face schooling in computer perception has been shifted toward engineering features rather than using the hand. Data is vital in obtaining better results from DNNs based on extensive and good quality data. For this purpose, a mechanism must be present for the training of data [11]. Facebook [12] and Google [13] provide extensive datasets from which a well-trained and good-quality model is obtained. The performance of deep learning highly depends upon the well-trained datasets.

In the field of computer vision and deep learning, research on deepfake generation and detection has seen rapid advancements recently. While a few review and survey articles have been published on this subject [3], [7], [10], [14], [15], [16], [17], [18], [19], [20]. However, they primarily focus on deepfake or synthetic samples generated using GANs, or are limited to analyzing biological features. Furthermore, these articles often overlook the critical aspect of data augmentation and are generally confined to studies published up until 2023. They also skipped over the introduction of advanced face manipulation tools and techniques as well as novel strategies for deepfake creation and detection. As a result, this article offers a concise but thorough summary from both theoretical and practical perspectives to provide the reader with a conceptual understanding and to encourage the development of new, more durable procedures. Therefore, the main contributions of this article are as follows:

- 1) A comprehensive overview of the deepfake generation process is provided, including a classification of deepfakes into four categories, their applications, and the current tools used for deepfake creation.
- 2) The most widely used deepfake detection techniques are summarized, with key features and limitations highlighted to facilitate further exploration.

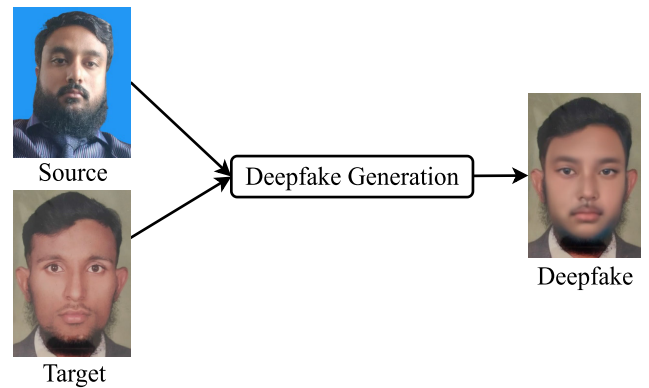


FIGURE 1. Media generated by Deepfake.

- 3) This review paper discusses in detail all currently available datasets utilized in deepfake technology.
- 4) Key points regarding the future potential of deepfake technology are identified and highlighted, based on the latest advancements in the field.

The remaining sections of this study are structured as follows: Section II delves into a detailed background of the study based on cited literature on the deepfake generation process, deepfake classification, deepfake detection process and applications of deepfake. Section III provides an in-depth overview of the research methodology employed, including research questions, research objectives, details on the search process (planning the review, search strategy, quality assessment criteria, and reporting the review), and the criteria for inclusion and exclusion of studies. Following this, Section IV investigates various visual based deepfake detection techniques. Section V systematically summarized the literature on deepfake detection datasets. Section VI explores potential concerns and future directions of deepfake technology. Section VII provides a conclusive summary of the study's findings and implications.

II. BACKGROUND

Deep learning is frequently used to discuss, analyze, and estimate the problem of detecting deepfake videos. Every video is categorized as either real or a deepfake. It is possible to create a contradiction of this type while creating and testing deepfake detection algorithms using videos that are neither authentic nor produced using deepfake generation methods. However, the problem becomes more complicated when the detection technique is applied in real-world situations. To focus more on deepfake, we discussed deepfake generation processes based on deepfake classification (synthetic face, face reenactment, facial attribute alterations, and face swap), deepfake detection process and deepfake beneficial and malicious applications in detail.

A. DEEPAKE GENERATION PROCESS

Tempering facial expressions or facial reciprocation as shown in Figure 2 are called deepfake [21]. One person's face

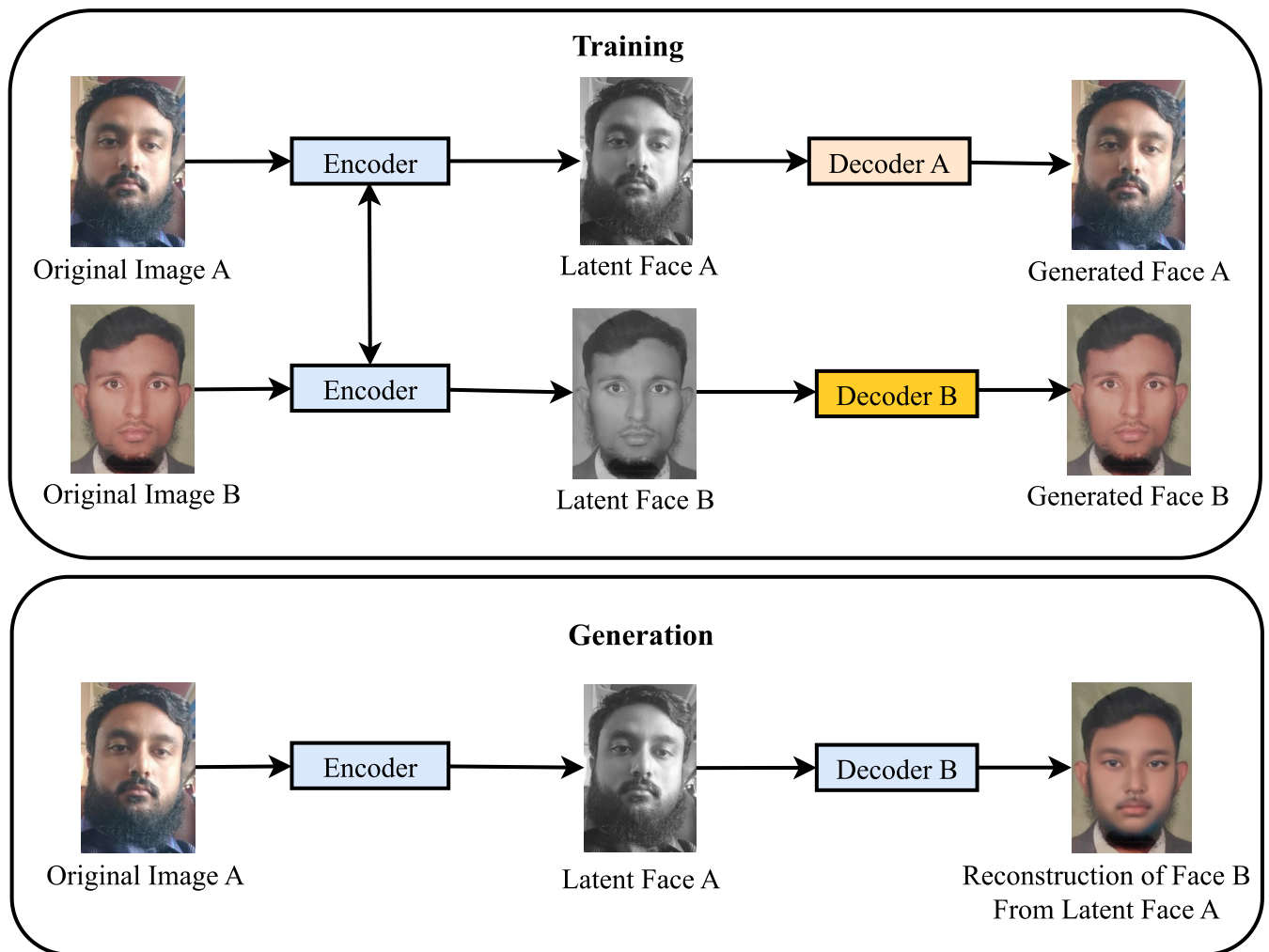


FIGURE 2. Deepfake generation process.

is swap with another to threaten the targeted person in deepfake videos, which may concern the public domain. Due to advancements in this technology, it is challenging to recognize realistic images and videos from fake images or videos because generated counterfeit images or videos resemble realistic ones [6]. Using AI, deepfake images or videos could be created by combining, replacing, merging, or imposing images or videos for scamming [22].

A Reddit user created the first deepfake, fakeApp, which utilized an autoencoder decoder pairing structure. The autoencoder extracts latent properties from face image data, which the decoder then reconstructs. Two encoder-decoder pairs are needed to switch the faces from source to destination. The same encoder networks share two pairs at this instance, and each pair shares encoder parameters with two network pairings, training on an image set. Using this method, the particular encoder may recognize and discover similarities between two pairs of facial photographs, which are frequently simple to identify due to similar features like the locations of the mouth, nose, and eyes.

After creating the deepfake video, two technological aspects are found in AI systems: 1) a generator and 2) a discriminator. The component that produces fake images or videos is called the generator; both mechanisms function together. Once the fake video has been generated, the discriminator evaluates it to decide whether it is real or fake. Even after it has been determined that the video is fake, the discriminator still provides the generator with advice on precautions to take and considerations to make while producing the following video. This leads to the creation of a network known as a Generative Adversarial Network (GAN) [23]. Firstly, the system must provide the necessary output type for the function of the GAN. After that, the system generates a dataset that the generator can use for training. The generator starts creating different kinds of videos as soon as it receives the necessary output, which is subsequently sent to the discriminator for analysis. The more errors discovered in fake videos, the more likely it is that the generator will be enhanced further for the next version.

A variety of tools are available which are easily accessible for the generation of deepfake; the deepfake generated by these tools is quite authentic that the media generated by them are not easily recognizable; in this regard, advanced deep learning and machine learning techniques are required for the deepfake detection. The list of deepfake generation tools is summarized in the Table 1.

B. DEEFAKE CLASSIFICATION

In this article, deepfake is classified into four main categories based on the facial manipulations they enabled: 1) Synthetic face, 2) Facial reenactment, 3) Facial attribute alterations, and 4) Face swap.

1) SYNTHETIC FACE

Synthetic face is the generation of an artificial face, that does not exist in real world. Synthetic face depends on discovering occluded illustrations of the facial datasets in order to produce a hyper-realist artificial face. The image's output is not actual since it was created without target features. Although the modeling and gaming industries benefit from this technology, an attacker may use it to steal someone else's identity. Yu et al. [48] introduced the Deep Convolutional Generative Adversarial Network (DCGAN), a more stable computational architecture, to increase training stability and also utilized deep convolutional networks instead of pooling and batch normalizing, and based on an arithmetic vector, they demonstrated an improved picture synthesis performance. A new network architecture called Progressively Growing Generative Adversarial Network (ProGAN) was released by National Vision Instrument and Advanced Graphics (NVIDIA) researchers a year later to improve end results' reliability and accuracy within the learning procedure. As an algorithm is trained, its characteristics gradually improve from poor-quality data.

StyleGAN is a kind of network derived from ProGAN [49]. After reviewing articles [50], [51], [52], it was observed that the researchers changed the generator methodology by utilizing Adaptive Instance Normalization (AdaIN) to handle the creator training at each CNN layer. In accordance with the provided vector, the creator usually synthesizes uniform posture or style. Additionally, researchers introduced random variation to the positioning of the synthetic person's pores, marks, beard, and hair. However, scientists discovered that there were noticeable water droplet-like anomalies in the artificial image as a result of StyleGAN's instance normalization. It is easily detectable due to its faint trace. In their parallel studies in the same years, they revised a new normalization strategy and released it as StyleGANv2 [53]. They were able to smoothly remove the artifacts and enhance the clarity of the image. Synthetic face databases are frequently created using ProGAN and StyleGAN [54].

2) FACE REENACTMENT

Reenactment, as opposed to face swap, is the transference of an individual's gaze or facial expression from source person to target while maintaining the identity of target source. This enabled the target to seem as though they are showing the expression or addressing things that they haven't said never. This method was widely used prior to the development of deepfake. The old methods obtain similar outcomes by using computer graphics. Blanz et al., for instance, used a 3D morphable model that was taken through the database of 3D scans to reproduce the input image [55]. Thies et al. allowed real time facial expression transmission by recording face performance using a common Red Green Blue (RGB)-depth sensor. They employed a mouth retrieval synthesis and a parametric model to alter the target's features to match the source expression in order to produce a high quality outcome [41], [56].

3) FACIAL ATTRIBUTE ALTERATIONS

Facial attribute alterations involve transforming some features of the face, including skin tone, eye color, age, hairstyles, and expression by preserving the target identity. It has the ability to change someone's look based on predetermined parameters. Star Generative Adversarial Network (StarGAN) was the exemplary domain to domain translation network for this technique [57]. In contrast to previously published studies [58], [59], [60], [61], by employing a mask vector approach, StarGAN facilitated multi domain training and emphasized style transformation across domains. Xiao et al. introduced elegant, a comparable multi attribute Cycle Generative Adversarial Network (CycleGAN) based translation network [62]. Researchers concentrated on learning models utilizing reconstruction loss, adversarial loss, and domain classification loss. Although the style transfer results were impressive, they are less acceptable than StarGAN since the modified outputs can contain unwanted artifacts.

4) FACE SWAP

Face swap entails changing faces from source to target while maintaining the facial expression of target person. A quick face swap technique was presented by Korshunova et al. in 2017 [63] to switch the identification of person A to B as long as the lighting setup, head position, and facial expression remain the same. Unlike a style transfer, where the content takes up the remaining space, this one describes the individual's style. The researchers employed modified MTCNNs with loss of content, style, and light function to transform the contents of A to B. They used 68 facial landmarks to align the resulting face using an affine transformation, and a segmentation mask was used to stitch the backdrop. Once trained to comprehend A's data, the neural network can have several targets, B, which enables a one to many face swapping method.

TABLE 1. Deepfake generation tools and key points.

Tools	Key Points
Faceswap [24]	Works on the pair encoder and decoder. Encoder shared publicly for deepfake.
Faceswap GAN [25]	Attention masks help with occlusion, artifacts, and better development of skin tone. Generates better resolutions up to 256x256.
Few Shot Face Translation [26]	Works on trained datasets for the extraction of embedded GAN images. Includes the semantic priors getting through the Spike Pattern Detection and Evaluation (SPADE) and Few-Shot UNsupervised Image-to-Image Translation (FUNIT).
DeepFaceLab [27]	A new version of the Faceswap adding new models H64, H128, LI-AEF128, and Single Layer Model (SAE). Help out numerous face pose modes such as Multiscale Convolutional Neural Network (MTCNN), Single Shot Scale-invariant Face Detector (S3FD), and Digital Library (Dlib).
DeepFaker (DFaker) [28]	Reconstruction of the face depends upon the Structural Similarity Index Measure (SSIM) function.
DeepFaketf [29]	An updated version of DFaker but implementation on test or flow.
AvatarMe [30]	Three Dimensional (3D) faces are reproduced from the images. Improvement of image resolution.
MarionETTE [31]	Does not require any other optional data for tuning up.
Deepfakes Web [32]	Allows the user to generate fake video online without installation of software and hardware.
DiscoFaceGAN [33]	Virtual facial development with different styles such as identity, pose, and gleam. Insert 3D antecedent into adversarial learning.
Reface [34]	Enable the users to replace their faces with those of celebrities or characters in videos and Graphics Interchange Format (GIFs).
StyleRig [35]	Generation of photorealistic portrait images, but lack of rig like over semantic face parameters. Manages data without requiring manual input.
Deepfake Studio [36]	Users may be able to integrate deepfake footage into larger projects by using Deepfake Studio's potential interoperability with other video editing programs. With this connection, the tool's adaptability in professional workflows is increased.
FFaceShifter (FSh) [37]	No further requirements of training. By configuring the target properties, face swapping is carried out with great realism.
DeepArt [38]	Allows to create unique pieces of art that blend their photos with the aesthetics of renowned artists like Van Gogh or Picasso.
Face Swapping GAN (FSGAN) [39]	Does not require specific training for face swapping and reenactment models. Helpful for expression and pose adjustment.
Style-based Generative Adversarial Network (StyleGAN) [40]	Based on the GANs architecture, new styles are developed.

TABLE 1. (Continued.) Deepfake generation tools and key points.

Face2Face (F2F) [41]	Reinvent the sequence of target video using F2F.
FaceMagic [42]	FaceMagic lets users easily exchange faces in images and videos while maintaining an appearance that the altered faces are authentically included in the original content. Sophisticated algorithms behind this function guarantee excellent outcomes.
FaceApp [43]	Allow the various transformations including change of gender, aging, and hairstyle.
Neural Textures [44]	It modifies the currently available video at real time rates in both static and dynamic modes.
Instagram-DeepFake-Bot [45]	Enable the user to generate the deepfake by using Instagram.
Transformable Bottleneck Networks [46]	The transformable bottleneck framework is used prior to applying spatial transformation in Convolutional Neural Networks (CNN) for image betterment.
“Do as I Do” Motion Transfer [47]	Uses in supervised learning, self automated transfer of the motion of video to the target video or person.
Neural Voice Puppetry [48]	Creation of a desired person’s photo realistic output video.

C. DEEFAKE DETECTION PROCESS

The primary concerns of deepfake detection are facial expressions, face alignment, and face classification. Facial features are the most fundamental step in deepfake detection. The performance of deepfake defections depends upon the excellent degree of official expression [64]. Identification of the face is a primary self regulating job in different applications, such as identity revivification systems, intelligent inspection systems, and automated immigration systems. Deepfake detection could be distributed into two branches: face identification and face verification. Face identification detects the most similar image from the given images. In contrast, face verification means that the given pair of images belong to the same or different pair of images [64], as shown in Figure 3.

Over the years, facial feature detection has become increasingly popular and stronger in research and practical applications. A well-known success in deep learning has encouraged facial feature reorganization. Conversely, facial wrapping relies on the expression, movement, and look of the face as well as the facade. In these cases, identifying a realistic face is a complex process. Gathering and classifying a large amount of data is a difficult and time-consuming operation. The data is also highly expensive and has fault tolerance; the publicly available datasets could not bear any variations. Face data augmentation comes into the cross to overcome the deficiency of facial training datasets. Face data augmentation is a technology that modifies or imitates virtual face patterns, as shown in Figure 4 to change gathered factual face patterns, increasing the data quantity.

Face data augmentation overcomes existing data limitations by enlarging or transforming the actual or simulated virtual face sample. Inadequate and limited data leads to accuracy and overwhelming inefficiency because of over-parameterization and overfitting problems. Face data augmentation plays a vital role in the performance of neural networks, such as 1) Desired features and attributes of faces could be obtained by applying controlling factors. (2) It is very cheap in this way that, with a little bit of transformation, it can get enlarged data as compared to the pileup and classify the actual data. (3) Artificially manufactured data has no bugs as compared to real-world data. (4) Data Augmentation has remarkable benefits like self-obstruction and equality of datasets with more intra-class variations [11]. Image augmentation could be achieved by flipping and rotating, cropping and resizing, color jittering, image warping, random erasing, and by adding noise patterns like as Gaussian, Rayleigh, Erlang (or gamma), exponential, uniform, impulse, salt and paper, speckle, poison, quantization, shot, and temporal noise.

D. APPLICATIONS OF DEEFAKE

Deepfake technology offers an extensive variety of uses that might be beneficial or malicious. The various applications of deepfake technology, positive and negative, are explained as shown in Figure 5.

1) BENEFICIAL APPLICATION

Deepfake has many beneficial and/or malicious impacts on daily life. Deepfake technology based on deep learning

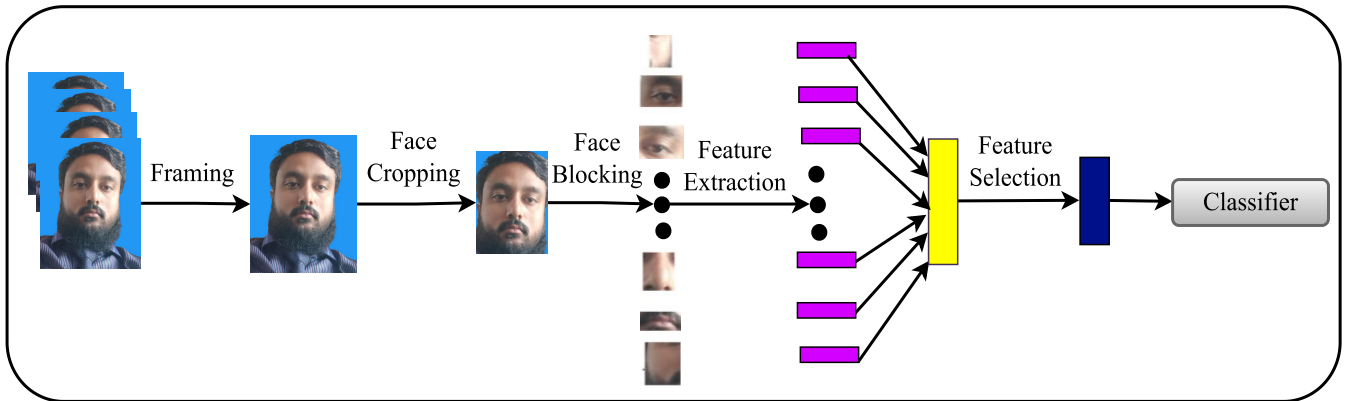


FIGURE 3. Deepfake detection process.

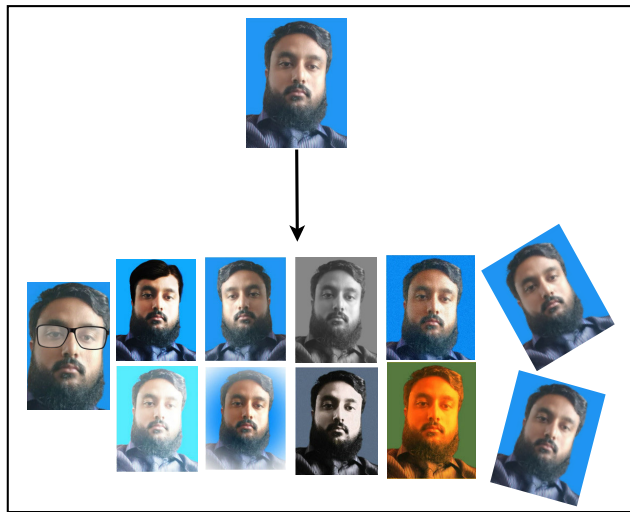


FIGURE 4. Augmented samples.

and machine learning is useful in several domains, such as material science, e-commerce, social media and the medical field, gaming and entertainment, cyberspace and education, and fashion as described in [65].

a: EDUCATION

The emergence of deepfake technology opens up new educational opportunities, for instance, language learners having real-world discussions or medical students practicing realistic surgical simulations. Deepfake has enormous potential to transform training scenarios and improve the learning process considerably. By facilitating virtual field trips and realistic reenactments of historical events, deepfake goes beyond simulations and allows students to experience learning outside the traditional classroom. The immersive nature of deepfake information may increase learners' memorability and engagement [66] deepfake can make remote meetings run more efficiently by interpreting speech and simultaneously modifying lip and facial motions to increase eye contact and

give the appearance that each individual is communicating the same language [67].

b: FILM INDUSTRY

Deepfake technology has the potential to benefit film companies in a number of ways. It can help, for instance, modernize film footage instead of reshooting it or creating synthetic voices for artists who have lost their own due to diseases. In the film industry, deepfake saves time and money by editing footage instead of reshooting them [68]. Additionally, it displays more outstanding entertainment performances, such as virtual cartoons, movies with visually appealing enchantments, and films dubbed in other languages [69]. Language barriers were broken by David Beckham in a 2019 worldwide malaria awareness campaign by appearing bilingual due to technology that changed his appearance and speech [70], [71].

c: ARTWORK

Concertgoers concentrate and have access to unique information with the deepfake by developing new artistic challenges. The Dali Museum in St. Petersburg, Florida, created a deepfake a few years ago to shake Salvador Dali's hand and learn more about his personality [72]. Virtual reality simulations of historical events or interactive instructional games could make better use of deepfakes.

d: INNOVATION

Deepfake produces visually appealing product dummies for retail clients. In the fashion sector, customers use deepfake to create virtual models for a better selection of clothing trends. Deepfake produces virtual news presenters and advertisements that have a realistic appearance. For instance, companies may hire glamorous models who are not actually glamorous models and use a range of models with various body types, heights, and skin tones in their fashion looks [73]. Furthermore, deepfakes allow users to create highly personalized material that serves as models. Along with allowing customers to try on items virtually before making

choices, the technology also creates personalized fashion advertisements that change based on spectators, the weather, and the time of day [74].

e: HEALTH CARE

Researchers are examining the utilization of GAN to identify anomalies in X-radiations (X-Rays) and their potential in order to accelerate progress in materials science and medicine. Deepfake allows the distribution of fictitious data for research purposes rather than patients' private information [75], [76]. The healthcare sector is positively impacted to a greater extent. Patients with Amyotrophic lateral sclerosis who create deepfake movies for perceptual comprehension have experienced more notable advancements using deepfake [77].

2) MALICIOUS APPLICATIONS

In human nature, he mostly tries to apply technology maliciously as compared to beneficially. Due to the different reasons that involve making deepfake content, such as making fun of someone, using it for taking an act of revenge, intimidation, face swapping of someone, and much more.

a: HARMFUL TO INDIVIDUALS AND ORGANIZATIONS

Deepfake not only impacts disinformation as are in per se, but it also has a lot of negative impacts if people remain in touch with misinformation for an extended period, including video, which means they cannot trust realistic information; that term is known as "information apocalypse" or "reality apathy." [78]. That is why people often dismiss original information. Deepfake techniques are also used for making pornography of famous personalities such as actresses [79]. Deepfake affects celebrities, politicians, and famous leaders because many photos and videos are available online for AI deepfake training. These types of deepfakes aim to ruin dignity and pranks with negative impacts on personality. Famous leaders like Barack Obama, Donald Trump, Nancy Pelosi, Angela Merkel, and even Facebook owner Mark Zuckerberg have faced deepfake issues [80].

b: HARMFUL TO SOCIETY

Deepfake affects government officials in visual data; they never utter the words, but these words are blamed on them. A fake viral video that a non-muslim man is celebrating in the mosque is seriously brutal against Islamic laws. Armed persons are shown to be fighting against their people in war and killing their own people badly. This act leads to societal war [77].

c: HARMFUL TO BUSINESS

Another negative impact of deepfake is cyber security. Deepfake effect on digital marketing and stock manipulation, such as the company's owner behaving bigoted or racist as a fake manager and showing false statements like profit and ruination as if committing a crime [81]. Deepfaked products

are used for blackmailing and stoppage of interest. Deepfake is a real-time digital parody of a chief executive officer of the United Kingdom (UK), for example, to make an urgent cash transition or give confidential information [82].

d: HARMFUL TO DIPLOMACY

Deepfake can disrupt national and international diplomatic relations; that harmful relation leads the nations toward war. Deepfake allows it to interrupt national security policies and democratic plans, and as a result, it violates civil norms. Deepfake is responsible for weakening national policies and procedures, effectively trapping the nations without a public war [77].

e: HARMFUL TO JOURNALISM

The ability to create deepfakes is increasing day by day, making it harder for journalists to confirm and distinguish between real news and events. By increasing the capacity of deepfake, it is challenging to recognize the realistic photos and videos from the fake. Such photos and videos lead to the proliferation of misinformation [83]. Verifying and recognizing the realistic contents of fake content is a burdensome and time-consuming task. The editor and the journalist spend more time on this burdensome task, casting news or reports in maximum average time and taking more workforce [84].

f: HARMFUL TO CYBER-SECURITY

Election related to deepfake interactions; weaken laws and regulations, erode public confidence, and cyber-threatening individuals and organizations. Deepfake encourages cyber-criminals to engage in online hoaxes [85]. Nowadays, deepfakes are working like masquerading and provoking innocent people to share their secret information for fraud. As a result, the ramifications of deepfake technology for society are expected to grow in the future years. It is critical to begin thinking about reducing the hazards of deepfake technology while maximizing its potential advantages. This will necessitate a collaborative strategy combining governments, technology corporations, academia, and civil society organizations.

III. METHODOLOGY

To process the systematic literature review, we focused on the survey [86], which had a more significant positive impact on the survey in software engineering. We divided this study into three phases (planning, conducting, and reporting) to classify the research review process, which was used to recognize, evaluate, and understand the different types of research works to strengthen our proposed Research Question (RQ)s. The data from the selected studies were extracted and monitored. We selected the five online digital repositories mentioned in Figure 6 to find the relevant data. Extractions of required data were collected based on RQs. The outcomes and method section defined which data type needed to be extracted. We considered the publication acknowledgement,

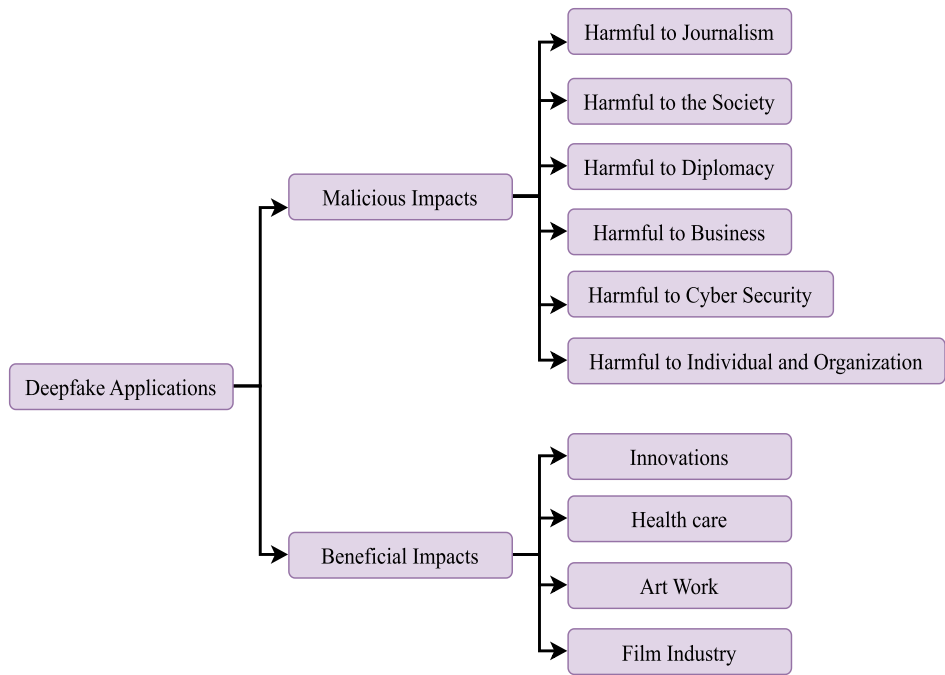


FIGURE 5. Categorization of Deepfake applications.

the author’s validity or information, publication area, and time in journals, workshops, and conferences. The different techniques were analyzed based on features used for deepfake detection. We also tried to find the research gaps among the different models, and the limitations were part of the literature review.

A. PLANNING THE REVIEW

The planning of this review involved identifying needs related to the purposes of writing a research article. The need for deepfake detection was related to the malicious impacts of deepfake and also focused on the existing research limitations in deepfake detection methods. We conducted a review consisting of the following stages, referred to in [87] and [88], that include search strategy, quality assessment criteria, and reporting the review.

B. SEARCH STRATEGY

The search strategy established relevant data collection norms. This phase aimed to collect and refine the relevant literature as much as possible. At this phase, it was decided where to start the search. Keywords and indexes were well-defined, as shown in Table 2.

1) POPULATION

To commence the review, we precisely defined the study’s population. The primary keyword for the research study was ‘deepfake’. This research review ensures a comprehensive exploration of diverse emergency scenarios impacted by

the technological transformation in the field of ‘Video Deepfake,’ and ‘Imagery Deepfake.’

2) METHODOLOGY OR TECHNIQUE

We used four feature representation domains for visual detection: spatial domain, temporal domain, spatio-temporal, and frequency base domain detection as demonstrated in Figure 7. For this purpose, to extract and monitor the data, different mechanisms were defined to obtain the desired information from the selected literature and also employed the keywords including ‘Deep Learning,’ ‘Machine Learning,’ ‘Utilizing the FakeCatcher,’ ‘Data Augmentation,’ and ‘Based on Feature Representations’ to strengthen the methodology. For data synthesis, our primary focus was to classify and recapitulate the results drawn from the selected literature. Empirical evidence was based on the following parts: 1) utilization of datasets in techniques, 2) key features mentioned in techniques, 3) models and methods used in different techniques, and 4) based on limitations described in techniques.

3) OUTCOMES

The possible outcomes of the research study mainly depend upon the methodology or techniques that were adopted for data collection and monitoring and had the impact of the outcome on research study applications. The keywords utilized to review the possible applications of the research study are as follows: ‘Face Forgery Detection,’ ‘Unconstrained Face Recognition,’ ‘Face Image Transformation,’ ‘Face Forensics,’ ‘Facial Images or Videos Restoration,’ ‘Detection of Face

TABLE 2. Search strategy for the selection of relevant research study.

Area	Keywords	Synonyms
Population	Deepfake	Deepfake OR Video Deepfake OR Imagery Deepfake.
Methodology	Deepfake Detection Techniques	Deep Learning OR Machine Learning OR utilizing FakeCatcher OR Data Augmentation OR Based on Feature Representations.
Outcomes	Deepfake Detection Applications	Face Forgery Detection OR Unconstrained Face Recognition OR Face Image Transformation OR Face Forensics OR Images or Videos Restoration OR Detection of Face Reenactment OR Detection of Face Swapping OR Detection of Face Synthesis OR Detection of Facial Edit Attributes OR Facial Manipulation OR Facial Expression Recognition OR Exploiting Visual Artifacts.

Reenactment,’ ‘Detection of Face Swapping,’ ‘Detection of Face Synthesis,’ ‘Detection of Facial Edit Attributes,’ ‘Detection of Facial Manipulation,’ ‘Facial Expression Recognition’ and ‘Exploiting Visual Artifacts.’

All possible combinations of related search queries (keywords or phrases) were used to get all data related to our RQs. Research queries were investigated using the boolean terminology (‘OR,’ ‘AND’) like as (Deepfake OR Video Deepfake OR Imagery Deepfake) AND (Deep Learning OR Machine Learning OR utilizing FakeCatcher OR Data Augmentation OR Based on Feature Representations) AND (Face Forgery Detection OR Unconstrained Face Recognition OR Face Image Transformation OR Face Forensics OR Images or Videos Restoration OR Detection of Face Reenactment OR Detection of Face Swapping OR Detection of Face Synthesis OR Detection of Facial Edit Attributes OR Facial Manipulation OR Facial Expression Recognition OR Exploiting Visual Artifacts). We used data repositories (shown in Figure 6) to collect relevant data from the period of 2017 to 2024.

We focused on RQs, which had a greater possibility of collecting as much relevant work as possible. The purpose and results of the research study depend upon choosing the appropriate questions, and it is the mandatory and foremost step in any research study. A meaningful research question leads the research study to the right and meaningful domain [88]. We raised the six primary questions for reviewing and analyzing the deepfake, which are the primary objective of the research study as demonstrated in Table 3.

To support the RQs, data extraction was made based on findings from the literature during the data extraction process, and these findings were synthesized and related to the associated and comparative bases. After analyzing the data, different tools (Histogram, pie chart, and Tables) and techniques were applied for better data visualization. We surveyed 2329 studies from different digital online repositories from 2017 to 2024 and focused only on the

most relevant studies (n=220) based on deepfake technology, as shown in Figure 6.

C. QUALITY ASSESSMENT CRITERIA

Regarding the study’s selection standards, we focused on the many selections (inclusion/exclusion) criteria points such as digital online repositories, languages, knowledge about authors, journals, magazines, technical reports, newsletters, Doctor of Philosophy (PhD) or other degree programs thesis for the requirements were considered. Properly adopting selection procedures led to pure and relevant literature to the RQs. We ensured that the literature findings were relevant, had no biases, and had well-defined selection criteria parameters. For this purpose, quality assessment plans were developed to measure the inclusion/exclusion criteria.

We consequently observed the excellence of the research work. Adopting poorly conducted studies led to results far from the desired outcomes. While writing the systematic review, these studies were pointed out or excluded. A refined categorization strengthened the evidence to analyze the data in a better way.

1) INCLUSION CRITERIA

- The title, abstract, or keywords were searched by the selected phrases.
- If the word ‘Deepfake’ was not mentioned in the keyword, title, or abstract but was defined in other parts of the literature, then this was part of our research study.
- Empirical study evidence was also included.
- Research carried out in 2017 or later.
- Research using machine learning techniques.
- Research using deep learning techniques.
- Study based on only visual deepfake.
- Selected abstracts, books, conferences, journals, keywords, papers, titles, and workshops were related to deepfake detection.

TABLE 3. RQs for reviewing and analyzing the Deepfake.

Sr No.	Research Questions	Research Objectives
RQ-1	What are the needs of deepfake detection?	To describe the beneficial and malicious impacts of deepfake.
RQ-2	What are the crucial domains for deepfake detection?	To discuss the techniques based on feature representation domains.
RQ-3	What are the key features utilized in various techniques for deepfake detection?	Discriminate the key features of different techniques.
RQ-4	What are the limitations of different techniques?	To define the limitations of different techniques.
RQ-5	What are the most common datasets used in deepfake detection?	To list down the datasets that are commonly used in deepfake detecting techniques.
RQ-6	What are the potentials of deepfake?	To describe the potential concerns and future directions of deepfake.

2) EXCLUSION CRITERIA

- Pieces of literature that were not in the English language.
- Our primary focus was on the images or videos temporization and some of the audio, whereas we excluded text temporization completely.
- We did not use magazines, master project documents, newsletters, patents, PhD thesis, or undergraduate or technical reports.

D. REPORTING THE REVIEW

Deepfake applications actually started in 2017, but no deepfake detection techniques were developed. In 2018, only nine articles on this topic were published in Science Direct. We included studies from 2018 and considered the research selection history up to April 2024. As of 2019, there were only 29 publications, but they rapidly increased from 2020 to upward. In 2022, the research work on deepfake detection was thrice as high as in the prior years, as shown in Figure 6. The thirst for deepfake detection increases as the malicious impacts of deepfake increase compared to the beneficial impacts of deepfake. The total number of studies from five online digital repositories based on deepfake detection is represented in Figure 6.

IV. VISUAL BASED DEEPAKE DETECTION TECHNIQUES

The process of communicating information or ideas using visual components, including images, image graphs, corresponding pair pixel position information, color, shape, texture, or in the video, including the inconsistencies between the frames with respect to space or time and other graphical forms is known as a visual representation. Based on the visual representation, deepfake detection involves the spatial, temporal, frequency and spatio-temporal domains, and different visual based techniques are used to explore these

domains, as shown in Figure 7 and most commonly used classifiers extracted from literature are depicted in Figure 8.

A. SPATIAL DOMAIN

The corresponding positions of pairs of pixels defined by their distance from one another and their orientation, which indicates where the second pixel is in relation to the first, is the spatial information taken into account. This pixel information is used for the detection of deepfakes. Using the pixel distribution in visual deepfakes could contribute to the spatial domain effects. The several kinds of spatial domain based detection techniques are covered in detail in this section as shown in Table 4.

1) FORENSIC BASED TECHNIQUES

Deepfake has several hints or marks that disturb the real samples, through which several methods could detect by focusing on the hidden landmarks and expressions. The Cozzolino et al. [89] claimed that as compared to conventional based techniques, it improves detection accuracy by emphasizing the use of rotation invariant characteristics that are computed broadly throughout the image. The PatchMatch algorithm, which enables quick approximate nearest-neighbor searches, is implemented by the authors to overcome the lengthy processing times characteristic of dense-field approaches. While preserving feature matching performance, this modification seeks to increase robustness against rotations and scale changes. According to the experimental results, the suggested method outperforms current dense-field techniques in terms of speed and robustness. The suggested approach may still be slower than some key-point based methods, especially in large images, even though it is faster than conventional dense-field techniques moreover a major issue with dense-field

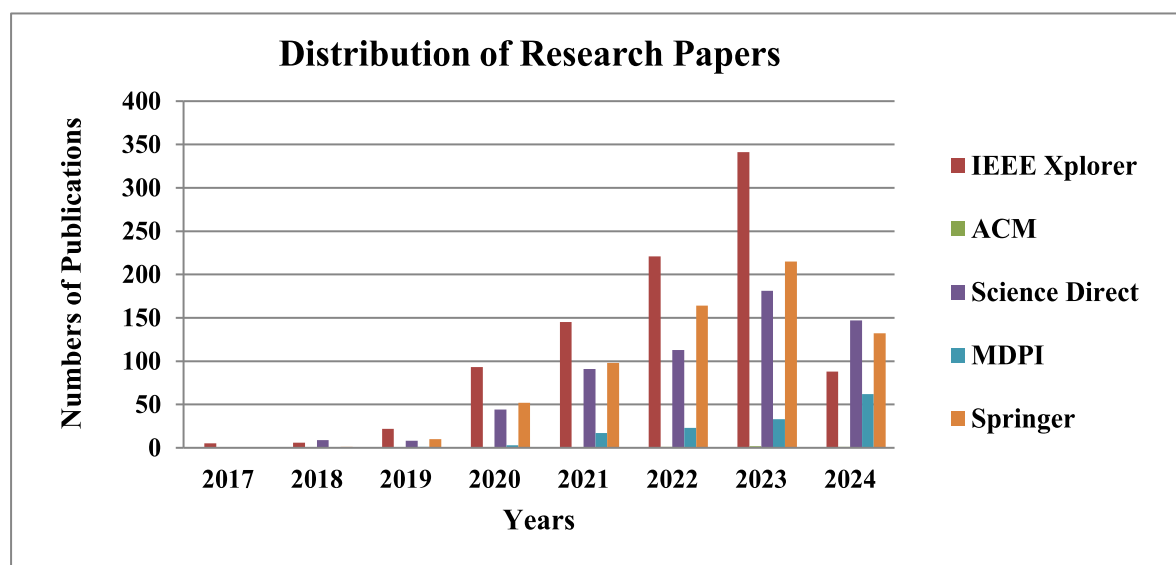


FIGURE 6. Numbers of studies with respect to years.

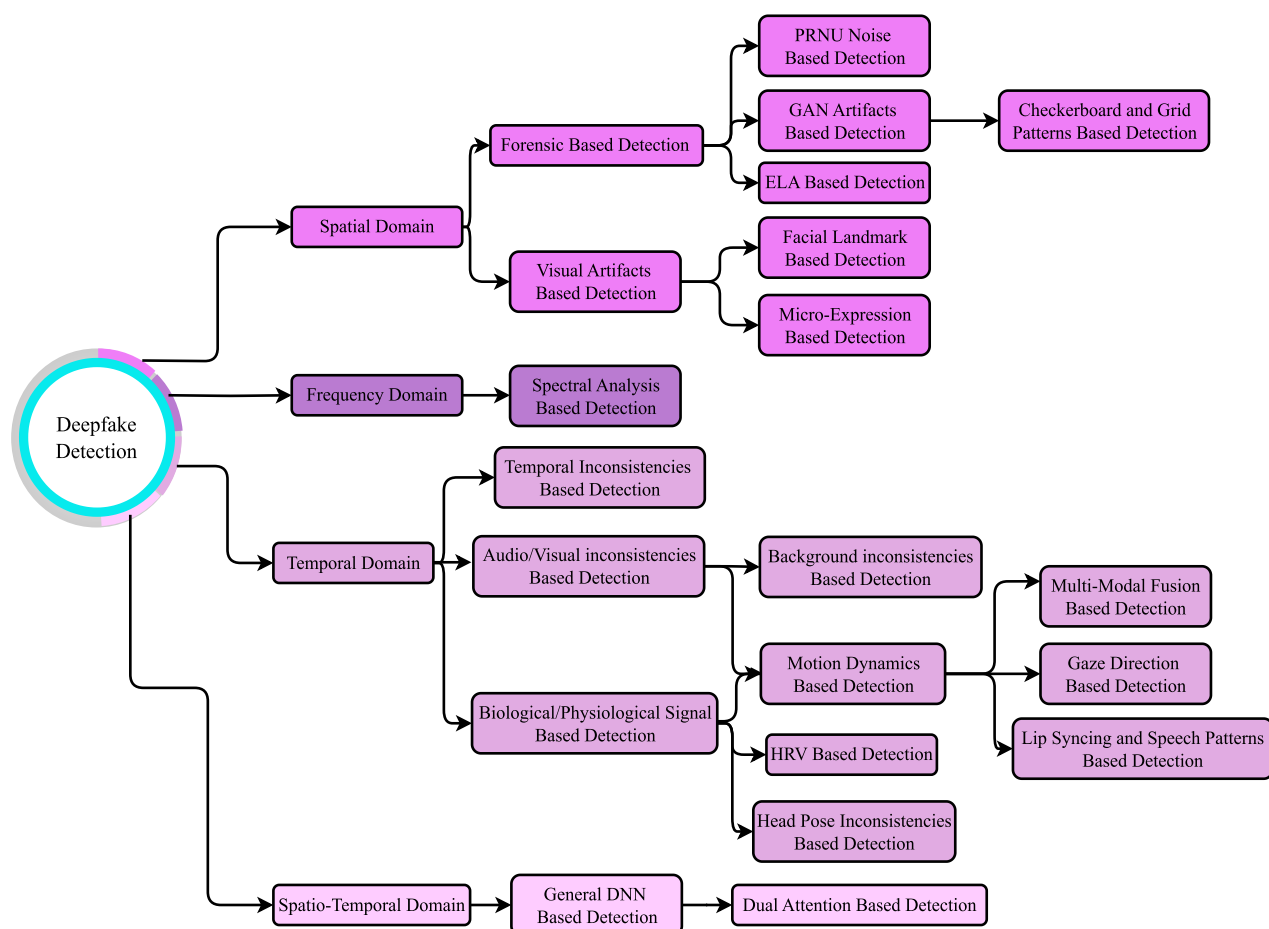


FIGURE 7. Deepfake detection techniques based on feature representations.

approaches is the algorithm's inability to detect copy-moves in smooth areas. Chen et al. [90] proposed a scheme that

focused on the features from the RGB and obtained noise vectors to collect the robust features. Real time assessment is

not necessary as a result of forensics. A forensic analyst may conduct other analyses along with a manual review during the examination.

a: PHOTO RESPONSE NON UNIFORMITY (PRNU) BASED DETECTION

Every digital image sensor introduces a signal that simulates noise into every captured image. This signal comes from differences in sensor pixels during the photoelectric conversion process. As a result, the PRNU can be seen as an inherent characteristic of each captured digital image. Koopman et al. [91] suggested the only Photo Response Non Uniformity (PRNU) based deepfake detection method. They demonstrated a considerable variance in the average scaled cross-correlation score between real and deepfake videos by comparing PRNU patterns of several frames within the same video. Their research, however, was restricted to a (private) self-made dataset with 16 deepfakes and 10 real videos.

Lugstein et al. [92] reported that PRNU based deepfake detection is a modification to earlier methods by Scherhag et al. [93] and Rathgeb et al. [94], which were utilized for the identification of different types of image manipulations, such as face morphing assaults on passport photos and facial retouching via beautifying apps. To detect deepfakes, they employed the same concept and utilized the suggested PRNU characteristics.

b: GAN ARTIFACTS BASED DETECTION

A great number of investigators took advantage of the detectable clues left by the inadequate design of GAN to investigate cases of fake detection. In order to train a classifier, Marra et al. [95] suggested utilizing raw pixels and traditional forensics features taken from real and artificial images. The co-occurrence matrix was utilized for the feature extractor by Nataraj et al. [96], indicated superior performance compared to classifiers trained using raw pixels on CycleGAN data. Guarnera et al. utilized the Residual Network Net (ResNet) models for classification tasks, which are based on the three hierarchical levels, 1: Distinguishes between real and GAN-generated images. 2: Differentiates between images generated by GAN and Diffusion Models and 3: Identifies specific AI architectures among the collected dataset while the study does not address the generalization of the classifier to unseen data. The updated version of **resent!** (**resent!**) model is ResNet-18 utilized by the Ahmed et al. [97] for the diabetic retinopathy to collect the artifacts from images. Tan et al. [98] suggested that the learning on gradients framework transforms images into gradients using a pre-trained CNN model, filtering out irrelevant content and retaining essential discriminative pixels for detection. The transformation model is fixed, allowing the gradients to serve as a generalized artifact representation, which reduces reliance on training data and enhances generalization but has the limitation that the proposed framework primarily focuses on detecting GAN-generated images. However, it has

limitations when it comes to detecting non-GAN models for deepfake detection.

It is frequently impossible to obtain the pre-trained GAN model used to create the artificial image in practical applications. Data augmentation techniques are used to get better results from the trained datasets. By applying a series of geometric and photometric transformations on training datasets also resolves the overfitting problems. However, different researchers utilized different techniques to overcome the problem of overfitting and made handsome approaches to achieve the generalization. Prajapati and Pollett [99] used augmentation and distraction to minimize the overfitting and achieve the generalization. The researcher focused on the Magnetic Resonance Imaging (MRI)s of faces based on SSIM while they relied on the GAN and effincNet_b0 for the feature extractions; due to the heavy augmentation and distractions, their result of the MRI-GAN method stepped down up to 74%. **Checkerboard and Grid Patterns Based Detection** based deepfake detection utilizes the unique artifacts frequently shown in fake images as a result of upsampling methods employed in GANs and other synthetic image techniques. This phenomenon is used as a forensic feature to detect content that has been tampered.

In order to detect GAN generated images, Li et al. [100] present GAN Artifact Similarity Estimation (GASE)-Net, a detection framework. A dataset that includes both pristine and GAN generated images is used to train the model. Every image is marked to be real or fake. Post-processing techniques like resizing and JPEG compression might mask GAN artifacts and impact detection performance. The impact of up-sampling methods, particularly up-convolution or transposed convolution, in generative CNN, such as GANs, is examined in the research article [101]. It draws attention to the fact that these techniques fail to replicate the spectral distributions of the natural training data. The authors suggest adding a novel spectral regularization term to the generator loss function to address the spectral inaccuracies and enhance the training stability of GANs. Although the spectral distribution problem is resolved in the paper, it admits that more theoretical research is necessary to comprehend the phenomenon's underlying causes and consequences completely.

c: ERROR LEVEL ANALYSIS (ELA) BASED DETECTION

Error Level Analysis (ELA) is an approach used in digital forensic analysis to identify anomalies in image compression that can point to manipulation or tampering. Although ELA was first developed to identify modification in static images, now it is utilized for deepfake detection, particularly when analyzing inconsistencies of compression effects across different areas of a video or image frame. Altered facial regions usually have different compression error levels than unaffected regions because deepfake videos usually have different re-encoded schemes for affected regions of the images or videos.

Rafique et al. [102] offered a model that used deep and machine learning techniques to categorize deepfake images effectively. Image alternations are detected by the ELA; after that CNNs to extract features, and Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) are employed to categorize the images. The model's accuracy fluctuates when the noise is added to the data. The high computational resources required to employ deep learning methods can sometimes be a barrier in real-time scenarios. Rafique et al. [103] proposed a model based on Alex Net and Shuffle Net with the ELA to differentiate between authentic and fake images. Despite the large dataset, the model's generalization to other datasets or real-world situations may still be constrained due to a limited number of images (2041). The study concentrated only on image-based deepfake identification, while the model failed to detect the deepfake in videos. The use of ELA methodologies can pose issues when dealing with multiple image formats or compression, and the proposed methodology by Rafique et al. [103] are incompatible in these situations. Geethanjali et al. applied the ELA to the images [104]. This technique highlighted the discrepancies in compression levels and presented the regions that require improvements. ELA served as a forensic tool to point out these discrepancies in the manipulated images. The model still performed poorly on unknown data despite utilizing dropout layers to reduce overfitting.

2) VISUAL ARTIFACTS BASED DETECTION

Due to processing power and generation time constraints, deepfake algorithms can only create face images of a specific size. They must go through affine warping to fit the source's facial arrangement. Face warping causes various artifacts as a result of the difference in resolution across the surrounding features and the distorted facial region. Realistic images are trained with the variational auto-encoder; this classifies them and other images treated as fake images, as proposed by Khalid et al. [105].

In order to identify face swapping techniques, Li et al. [106] identified the blending boundaries; a second approach involves using neural networks, such as customized deep networks [107], traditional neural networks and so on, to capture the forged features. The efficiency of the neural approaches was incredibly great [108]. However, the reliance on training datasets substantially restricts the ability to generalize the model, which is crucial for practical use.

Hong et al. [109] suggested a technique in which every frame in a real video should have a consistent fingerprint or artifacts. On the other hand, deepfakes inherently generates inconsistent artifacts due to the substituted face areas. In order to identify the discrepancy from the generated data, they suggested a self-supervised deepfake identification technique that uses an Artifact Discrepant Data Generator (ADDG) and a Deepfake Artifact Discrepancy Detector (DADD). The ADDG generates fake samples by altering the facial area using conventional processing tools, utilizing only real video

frames. The DADD uses a multi-task learning method, links each sub-task to a particular kind of generated data, and integrates the sub-tasks to achieve the final results. These methods are beneficial as the visual artifacts are adequately defined.

a: FACIAL LANDMARK BASED DETECTION

In deepfake detection, facial landmark analysis analyses the stability and coherence of facial features (eyes, nose, eyebrows, chin, mouth, chin and jawline) to determine differences between synthetic and real data. The method utilizes distinct spatial relationships, landmark consistency, and feature symmetry to improve detection accuracy. In the dlib library, 68 (x, y)-coordinates corresponding to facial features on the face are estimated utilizing a trained facial landmark detector [63].

Yang et al. [110] presented a new technique to identify deepfakes by examining variations in head positions that are inferred from facial landmarks. It draws attention to the fact that deepfakes frequently provide misaligned landmark positions, which causes notable variations in predicted head poses. The study acknowledges that head posture analysis can be used to identify landmark location inconsistencies, even if they might not be visually noticeable. In situations where landmark detection is inadequate, landmark accuracy could reduce the model's efficacy. The author Li et al. [111] study is based on a novel image representation known as face X-ray. As a typical face transformation technique, this representation helped determine whether an input face image is a blend of two distinct images. The framework depends on the blending phase rather than particular artifacts from manipulation techniques, improving its applicability in various situations. When working with low-resolution images, performance decreases because the fake evidence is less noticeable. The current arms race between fake creation and detection is a typical problem in forgery detection. Liao et al. [112] presented a novel work based on the Facial Muscle Motion (FAMM) for detecting deepfake social short videos such as TikTok or Facebook reels. They focused on continual facial landmarks taken from video frames; the FAMM framework creates facial muscle motion features. The method aids in spotting strange facial expressions that are a sign of deepfakes. In this approach, a new method of video forensics was adopted called the Dempster-Shafer theory, which combines various forensic expertise to improve detection accuracy. The authors claimed that the model have good performance on FaceForensics++ (FF++) dataset but not for others. The study admits that the various coding techniques social networks employ to compress videos could provide new difficulties that the studies did not fully explore.

b: MICRO-EXPRESSION BASED DETECTION

Finding minute, uncontrollable facial emotions that usually last less than 0.5 seconds is the primary goal of micro-expression analysis in deepfake detection. Due to their

transient nature and the sensitivity of the facial muscle movements involved, these micro-expressions, such as small lip movements, eye twitches, or eyebrow raises, are difficult to replicate in deepfake movies. These spontaneous facial expressions in real individuals frequently reflect their emotional states. However, deepfake algorithms frequently fail to precisely replicate these complex emotions, leading to observable anomalies or a deficiency of predicted micro-expressions.

Yildirim et al. [113] highlighted the significance of nonverbal features, particularly micro-expressions, for identifying emotion signals during interactions with others. The study used a deep learning algorithm to evaluate micro-expressions with an accuracy of 74.17%. A model is trained using the Facial Expression Recognition 2013 dataset to categorize forgeries data. The dependence impacts the accuracy of the results on the Facial Expression Recognition 2013 dataset, which might not accurately reflect the diversity of micro-expressions in real-life scenarios.

B. FREQUENCY DOMAIN BASED DETECTION

Frequency is the rate of change in an image. High frequencies are seen in areas of the image where the color changes rapidly, like sharp edges, while low frequencies are found in areas where the color changes gradually, such as broad surfaces with solid color. The real images have a higher frequency distribution than the fake images, and these variations are helpful in the detection of deepfakes as shown in Table 5. Durall et al. [142] are the first research group to focus on the Discrete Fourier Transform (DFT) to detect face manipulation. They found that the deepfake information has been detected by utilizing the average amplitude of each frequency band Discrete Cosine Transform (DCT) is used by F3-Net [143] to retrieve frequency domain data and examine the computational aspects of face forgery detection. The color combination in GAN images and the type of camera used this information to design the network, utilizing the prior knowledge proposed by McCloskey and Albright [144].

GAN generated images are explored using the frequency spectrum discussed by Frank et al. [145]. While presenting a modification of the localization architecture utilizing spatial maps and frequency domain association, Jeong et al. [146] recommends a GAN specific identification using artificial fingerprints in the frequency domain. GAN generated images have different frequency domain distributions from real images. By adopting the distribution of frequency domains, it could predict the deepfake because the primary characteristic of Frequency-level Perturbation GAN (FrePGAN) is ability to produce frequency level perturbation maps, which aid in blurring the boundary between faked and original photos. Enhancing the generalization of deepfake detectors across different GAN models is based upon this. The frequency perturbation of FrePGAN and pixels distribution are the two primary parts of the model. Together, these modules improve detection capabilities by concentrating on irregularities across the pixel and frequency levels.

The tendency of detectors to overfit to training conditions because of the distinctive appearances of frequency level artifacts which can differ depending on CNN architectures and training categories is one of the major drawbacks that has been found. Wang et al. [147] proposed that deepfake detection can extract richer attributes according to the suggested strategy, which combines information from the RGB and frequency domains. The model exhibits enhanced robustness, especially in identifying low-quality compressed deepfake images. The technique performs less well on datasets such as F2F and FaceSwap, perhaps because expression tampering techniques produce smaller tampered regions. The concentration on particular regions (such as the lip) in circumstances such as F2F fraud reduces the number of observable forger clues, which affects the overall detection accuracy. Yu et al. [148] proposed a scheme to detect a distinct GAN uniform fingerprint that preserves various frequencies and patches applied on generated images by a neural network.

The method suggested by Guo et al. [149] concurrently captures information in the spatial and frequency domains to effectively describe manipulation hints provided by Deepfakes. By using both methods, it is possible to prevent the loss of crucial forging hints that might happen when using solely frequency-domain characteristics. one of Space Frequency Interactive Convolution (SFICConv)'s components that uses three stages generation, interaction, and fusion to learn space-frequency characteristics. The model can now identify even more minute alteration. The study observes that the generalization power of SFICConv may be weakened by the lack of common characteristics between datasets such as FF++ and DFDC. When models trained on one dataset are tested on another, this becomes more clear. SFICConv has limitations in its usefulness across different datasets, even if it typically increases detection accuracy. It also does not always yield the optimal results in every case.

1) SPECTRAL ANALYSIS BASED DETECTION

A potent technique for deepfake detection is spectral analysis, which employs frequency-domain data to identify irregularities in images or video frames that indicate manipulated or fake content. Techniques for spectral analysis, which convert data into the spectral (frequency) domain, draw attention to particular patterns and artifacts suggestive of deepfakes, which typically differ from others. The component of spectral analysis is Fourier Transform (FT) and DCT. The FT highlighted these frequencies and used them for deepfake detection, which is very difficult to detect using spatial or temporal methods. DCT is a widely used method in digital image processing that has shown effective results in deepfake detection by analyzing compression errors and frequency variations. DCT based techniques generate synthetic data, identify discrepancies, and collect spatial frequency information by decomposing image or video data into cosine components. CNNs and DFT analysis are the main methods employed by Alkishri and Al-Bahri [150] to detect

TABLE 4. Summary of Deepfake detection methods based on spatial features.

Classifier/Technique	Key Features	Dealing	Dataset	Ref	Year	Limitations
SVM, Random Forest (RF), Multi-Layer Perceptron (MLP), and shallow	Discriminate features are extracted with the help of the bag-words method, and RF, MLP, and SVM use the characteristics for further classification.	Images	Labeled Faces in the Wild (LFW) face	[114]	2017	Need attention to the live videos.
Shallow, XceptionNet, and Handcrafted	Comparing the performance of multiple image forgery detectors under ideal and compromised settings against image-to-image translation attacks with the use of regular compression on social media platforms.	Videos	36,000 or 256×256 color images, providing a diverse and extensive set of data	[95]	2018	During compression, some information crucial for detecting fake images is lost and cannot be recovered.
PRNU	Focuses on the alliteration of the light-sensitive sensor of digital cameras and the reason for their imperfections.	Videos	10 authentic and 16 deepfake with 20 to 40 second duration	[91]	2018	The dataset is too small.
CNN and Mesoscopic Net (MesoNet)	Helpful in deepfake detection for videos at the mesoscopic analysis level.	Videos	FaceForensics	[107]	2018	Dataset are complex.
Encoder-Decoder	Embeds space dispersion Features	Images/ Videos	FaceForensics and CelebA-High Quality (HQ)	[115]	2018	Requires large features.
PRNU	Employs spectral characteristics to measure changes brought forth by morphing and spatial features to analyze PRNU value distributions.	Images	Dresden image database and flatfield	[93]	2019	It can be required to combine numerous features to improve performance.
CycleGAN and StarGAN	Identifies GAN manipulated images and contrasting its methodology with current steganalysis and deep learning based technique.	Images	CycleGAN dataset and StarGAN dataset	[96]	2019	Localizes manipulated pixels in GAN generated fake images to further enhance detection capabilities.
Visual Artifacts and CNN	Deforms the artifacts of the amalgamated methods.	Videos	UADFV and TIMIT	[106]	2019	Improves the robustness of detection concerning multiple video compression.
Physiological Signal and SVM	By using the head pose, landmarks are detected for the detection of deepfake.	Videos	UADFV and DARPA MediFor GAN	[110]	2019	Low performance on blurred images.
CNNs, Adaptive Boosting (AdaBoost), and Extreme Gradient Boost (XG-Boost)	Identifies disturbed face regions	Images	Faceswap and SwapMe	[116]	2019	Do not work for entire image deepfake detection. Only for RGB schemes.
SVM and Forensics	Determines the inconsistencies between the different color combinations as Hue, Saturation, and Value (HSV) and Luma (Brightness), Luma-Blue, and Red-Luma (YCbCr) hues are present in the residual field.	Images	CelebA, LFW, Flickr Faces High Quality (FFHQ), and GAN generated images,	[117]	2020	Improves the performance in practical detection situations.
KNN, SVM, and Latent Dirichlet Allocation (LDA)	Expectation Maximization (EM) algorithms can be used for feature extraction and better work on realistic images of people affected by GAN.	Images	CelebA, AttGAN, Group-Wise Deep Whitening-and-Coloring Transformation (GDWCT), StarGAN, StyleGAN, and StyleGAN2	[118]	2020	No work on the standard image editing and malicious processing robustness.
Visual Geometry Group (VGG) and ResNet	With the development of adversarial perturbations advancement in deepfake detection	Images	CelebA and Few-Shot face translation GAN	[119]	2020	It takes so much time even to process a single image.
Visual Artifacts and CNNs	Works on the face-wrapping manipulation.	Image/ Videos	DeepFake Detection Challenges (DFDC), Celeb-DF, and FF++	[111]	2020	Does not identify the completely fake images.
Attribution based model and ResNet50 model while pre-trained on VGGFace2	No supervision data is required for the detection of deepfake videos based on ABC metric	Videos	DeepfakeTIMIT, Youtube videos and COHFACE	[120]	2020	To get better performance, more work on physiological signals is needed.

TABLE 4. (Continued.) Summary of Deepfake detection methods based on spatial features.

Long Short-Term Memory (LSTM) and CNNs	Frames and inter-frames structural inconsistencies	Videos	FF++ and YouTube videos	[121]	2020	Low performance on compressed videos. Not effective for frame level discrepancies.
One class autoencoder	Embedding spaces are helpful for the determination of anomalies.	Videos	FF++	[105]	2020	For more generalization, there is a need to be more robust in the one-class autoencoder.
DADD	Determines the difference between deepfakes and pristine videos by concentrating on artifact differences brought about by frame modifications, providing a self-supervised method.	Videos	FaceForensics and Celeb-DF	[109]	2021	Low performance on blur videos.
PRNU and SVM	PRNU characteristics are explored as spectral and spatial attributes.	Videos	DeepfakeTIMIT and FF++	[92]	2021	Has not performed sufficiently well in deep learning methods.
ResNet50	Detect the face forgery and identify the manipulated facial regions.	Videos	FF++	[90]	2021	Do not work on unseen face manipulation.
CNNs	Focuses on the slight texture discrimination of the image conspicuous.	Videos	FF++,	[122]	2021	Calculations are expensive and have no effective performance for untrained data.
CNNs	Circular symmetrical discrimination faces.	Images/Videos	DF-TIMIT, FF++, DFDC, and Celeb-DF	[123]	2021	It does not work for entirely unrealistic data.
Encoder-decoder	Discrimination of multi-level consistencies.	Videos	FF++ and DFDC,	[124]	2021	Cannot operate on unseen data.
Attention Mechanism with CNNs	To collect the subtle artifacts, different local regions are used as multiple attention maps and texture features.	Videos	Celeb-DF, DFDC, DF, and FF++	[125]	2021	Works for highly sensitive data that is restricted to the spatial domain.
CNNs	Works for multi-domain users to explore better features.	Videos	DFDC, Celeb-Df, and FaceForensics	[126]	2021	It is not better for a while when talking head poses.
MRI-GAN with SSIM	Uses the SSIM to measure real and fake differences. Augmentation and text or shape distraction are applied to achieve the generalization.	Videos	DFDC, Celebe, FDF, and FFHQ	[99]	2022	Due to heavy augmentation accuracy disturbance. Very initial SSIM and EfficientNet_b0 are used. No setup for blur videos.
Vision Transformers (ViT) Model	Creates a multi-stream network by utilizing the ViT model to detect deepfake.	Videos	FF++	[127]	2022	No mechanism for low-quality images and social media content.
Three Dimensional Convolutional Neural Network (3DCNN)	Introduced the portrait image generator. Allowed semantically guided face editing in 3D.	Videos	CelebAMask-HQ and FFHQ	[128]	2022	The issue of real-time free view image editing remains unresolved.
Fusing human examiner decisions	Most effective identification accuracy is found in a weighted fusion that accounts for the examiners' confidence in their opinions.	Videos	Face Recognition Technology (FERET)	[129]	2022	Human examinations fluctuate.
Identity aware dynamic network	Builds an Identification-aware Dynamic Network (IDN) by utilizing the Identity Injection Module (IIM), which focuses on the weights prediction and weights modulation for more efficient identification information injection.	Videos	FF++ and VGGFace2	[130]	2022	Only applicable for face swapping.
GAN and 3D-shape model	Discovers the directions in latent space of the pertained StyleGAN2 model by using the VoxCeleb1 dataset; on this basis, deepfakes are detected.	Videos	VoxCeleb1	[131]	2022	There are some visual artifacts in the reenacted photos at extreme source and target positions.
CNNs	Builds the CNN framework using cross-band co-occurrence matrices to extract discriminant features for the real.	Images	Validating Synthetic Image Detection and Source Linking Methods on a Large Scale (VIPPrint) and StyleGAN2	[132]	2022	No approach for both a black-box and a white-box situation in which the band associations are manipulated deliberately to fool the detectors.

TABLE 4. (Continued.) Summary of Deepfake detection methods based on spatial features.

CNN and GocNet	Uses the TP module, which enables the classical gradient operator in CNN to learn the manipulation traces. Also utilizes the MTA for feature refinement with a gradient operator to capture the altered traces. Finally, GocNet is used by combining the TP and MTA modules for deepfake detection	Videos	FF++, DFDC, High Frequency Fourier (HFF), and Celeb-DF	[133]	2023	Requires heavy computational resources. Time consuming.
Convey Brain Oversight to Raise Generalization (CYBORG)	Focuses on perceptual intelligence in relation to loss function to concentrate on human judgment regarding salient regions.	Images/ Videos	CelebA-HQ, FFHQ, Face Recognition Grand Challenge (FRGC)-Subset, Synthesis of Realistic Example Face Images (SREFI), ProGAN, StyleGAN, and StarGANv2	[134]	2023	The only saliency maps used are those linked to practical human judgments.
GAN	Suggests a brand-new method called deep notch that reduces artifact patterns in fake images without losing the quality of the image by utilizing the implicit spatial-domain notch filtering.	Videos	CelebA, ProGAN, Spectral Normalization GAN (SNGAN), Moment Matching GAN (MMDGAN), and CramerGAN	[135]	2023	Misclassification of images based on a predefined decision line, indicating the need for further advancements in detection techniques.
CNN	Uses dynamic graph learning to apply relation aware features in the spatial domains	Videos	WildDeepfake, FF++, DFDC, DF-v1.0, and Celeb-DF	[136]	2023	The quality of the input data and the complexity of the modification techniques might affect efficiency.
MesoNet, Face Warp Artifacts (FWA), XceptionNets, and capsule networks	Recognize images with different resolutions.	Videos	DF-Platter	[137]	2023	Higher end computing mechanism is needed for training.
GAN	To manage poor quality data, suggest a High Frequency Enhancement (HiFE) network.	Videos	FF++, Celeb-DF, and OpenForensics	[138]	2024	Does not focus on the temporal domain.
CNN	Two benchmarks, DiffusionDB-Face and JourneyDB-Face, are designed for deepfake detection. Proposed that the capabilities of these benchmarks are greater than those of the existing ones.	Images	JourneyDB and DiffusionDB	[139]	2024	Images are either too realistic or too cartoonish. A noticeable discrepancy in the gender distribution that emanates from the original datasets.
GAN	Forgery detection is achieved by integrating spatial domain data within a knowledge distillation framework.	Videos	FF++ and Celeb-DF	[140]	2024	Overfitting is identified as a probable cause for the model's lower performance on the F2F dataset compared to the backbone network Xception.
CNN	Forgery Trace Generation Network (FTG) is responsible for extracting inconsistent forgery traces from frequency and spatial domains, while the Forgery Trace Discrimination Network (FTD) is used for classifying real and fake videos based on the extracted forgery traces.	Videos	Celeb-DF, DFDC, and FF++	[141]	2024	Challenges related to scalability, generalization to new types of forgeries.

deepfake images. By examining high-frequency features of images, the DFT enhanced the process of identifying artifacts produced by GANs. Then, the images are processed by two classifiers: VGG16 and Dense Net-121. Ahmed et al. [151] utilized the same techniques VGG16 to explored the kidney stone artificial images. With 80% of the dataset going toward training and 20% toward testing, the dataset is divided into training and testing sets. The approach comprised an extensive evaluation of the classifiers' ability to identify fake images, with an emphasis on the variations

in frequency spectra between fake and real images. The study [150] indicated that although the suggested techniques work well, they might be improved, especially in terms of modifying the strategy better to handle low, medium, and high-quality data. The research article [152] focused on a hybrid method to identify deepfake facial modifications by combining GANs and CNNs. HFF modes of real and GAN generated images are analyzed, and the results show that GAN generated images consistently fail to reproduce high-frequency features. The technique involved eliminating

GAN fingerprint artifacts from frequency-domain modified images.

The Amin et al. [153] utilized a frequency spectrum analysis to identify deepfake frames by investigating the spectral abnormalities and statistical characteristics in the image's color channels. The model attempts to distinguish between authentic and fake images and is influenced by conventional image forensics. Both supervised and unsupervised learning strategies are used in the process. Unsupervised learning utilizes the GMM(FSSF)-EM technique to differentiate between actual and fake faces, while supervised learning trains models on particular datasets. Because there are no correlation constraints between color channels when generative models generate images, the authors find that they leave spectral artifacts in the frequency domain. The study noted that further research should examine other features or approaches to improve detection skills, as the emphasis on spectral features might not capture all aspects of deepfake detection. The AUthenticates SOcial Media images (AUSOME)-2 system, which used machine learning techniques and frequency analysis technologies to authenticate social media images, is presented by the Poredi et al. [154]. In order to extract characteristics and fingerprints from images in the frequency domain, the methodology focused on spectral analysis techniques, particularly DCT and DFT. The AUSOME-2 model can learn the properties of images produced by various platforms because it is trained and verified using a broad dataset. Although the article emphasized the efficacy of the AUSOME-2 system, it doesn't go into great length about any potential drawbacks or difficulties encountered throughout the research. The research does not address the computational resources required to train the model, which might pose a hurdle for some users.

C. TEMPORAL DOMAIN BASED DETECTION

Deepfakes are also generated based on temporal information. Temporal information could be defined as videos comprising a series of visual frames, while these frames include inconsistencies or chronological data. Frames of a video change concerning time; this information is known as temporal information. Inconsistencies and chronological data provide an excellent source for detecting deepfakes, as discussed in Table 6.

1) TEMPORAL INCONSISTENCIES BASED DETECTION

Flickering between adjacent styled frames and uneven stylization of moving objects are two visual cues that indicate temporal inconsistencies [162]. For deepfake detection videos, temporal inconsistencies between the video frames provide vital cues. Recurrent Neural Network (RNN)s or other similar networks are used to determine these inconsistencies.

Gu et al. [163] provide a novel framework for deepfake video detection called region-aware temporal inconsistency

learning to enable various regions to acquire self-unique temporal inconsistency. This system automatically creates relevant temporal filters. The Cross-Snippet (CS) attention module enables the CS information to deal with deeper representations when coupled with the specifically created sampling technique. A temporal aware model was proposed Guera et al. [164] to reorganize deepfake videos; a CNN was used for feature-level extractions from frames. After this, these extracted feature frames were sent to the LSTM layer [165] to check the manipulation sequences between the frames. At last, videos were organized as fake or real with the help of the softmax function. We can determine the per-pixel motion behavior called optical mechanism flow from the contagious frames. Gu et al. [166] presented a system that combines the temporal feature analysis skills of LSTM with the feature extraction capabilities of CNN. This framework is based on the extraction of temporal inconsistencies. Hosier et al. [167] proposed that video temporal points could be detected from the speed manipulation, and re-encoded frames predict the deleted or newly added frames between the videos.

2) AUDIO/VISUAL INCONSISTENCIES BASED DETECTION

In the past, linguistic expertise regarded as crucial domain knowledge has been widely used to deal with issues with audio visual consistency [168]. The researchers of [169] suggest a Siamese network-based method to distinguish between authentic and fraudulent videos by comparing multimedia and emotion-based variations in speech patterns and face movements. In addition, Chung et al. suggested [170] utilizing the 3DCNN for feature extraction in space and time domains. The research discussed in [171] combined visual and audio characteristics as inputs for LSTM networks for examining speech consistency, but it was ineffective in detecting deepfakes [21]. Deepfake is detected using visual artifacts such as mouth area and audio associated with lip-syn. Agarwal et al. [172] notify in their proposed scheme that deepfake video phonemes (spoken words) do not match with the visemes (outer region of mouth). This method was only related to the Barack Obama deepfake videos. While pronouncing these words M, B, and P, the visemes would ultimately close. The audio and visual aids have a precise correlation with one another. Awareness of Boundaries, a multimodal method called temporal forgery detection, was presented by Cai et al. [173] to accurately estimate the boundaries of fake sequences using audio visual data.

While an audio encoder using a Two Dimensional (2D)CNN learns features gathered from the audio, and video encoder using a 3DCNN learns frame level spatial information. Yang W. et al. [174] also leveraged differences between audio visual components to detect deepfakes. A cross-model classifier cooperated for tampering detection works with a multimodal joint-decoder to simultaneously achieve multimodal interactions and combine audio and visual information.

TABLE 5. Summary of Deepfake detection methods based on frequency.

Classifier/ Technique	Key Features	Dealing	Dataset	Ref	Year	Limitations
Intensity Noise Histograms (INH)	Analyze color cues and ability to discriminate between actual and GAN generated images.	Images	Media Forensics Challenge 2018	[144]	2018	Does not investigate the effects of outside variables such as post-processing methods, compression artifacts, or image resolution. Reliance on a relatively small set of training data.
Forensics (GAN artifact) and CNN	Presents the GAN fingerprints related to the image across different frequencies.	Images	CelebA, Large-scale Scene UNDERstanding Challenge (LSUN)	[148]	2019	Limits performance of post-processing operations.
Forensics(GAN artifact) SVM	A comparatively uncomplicated forensic depends on the frequency of underexposed pixels gives significant differences between the camera imager and GAN generation.	Images	GAN crop, GAN full, Media Forensics Challenge and CelebA HQ	[155]	2019	Does not work on the complete band of GAN generated images.
SVM and KVM	utilizing a classical frequency domain analysis followed by a basic classifier.	Videos	Faces-HQ, CelebA, and FF++	[142]	2019	Some samples are misclassified due to overlapping standard deviations.
Lifelong Few-Shot (LFS)	Effectiveness of frequency-aware face forgery detection by mining small-scale, detailed artifacts, particularly beneficial in low-quality videos	Videos	FF++	[143]	2020	Only focus on the spatial information.
SVM, LDA, and KNN	Developing the expectation maximization technique to improve feature identification outcomes using a fingerprint that displays convolutional evidence while creating GAN images.	Images	FF++ and GAN generated images	[156]	2020	Forensics data of real images are not included.
Siamese network	To identify deepfakes, voice and facial embedding vectors are extracted. Deepfake could be extracted based on motions such as face in the speech.	Images	DeepfakeTIMIT and DFDC	[157]	2020	No Works for video frames. Needed more artifacts.
Shallow and CNN	Artifacts in the frequency domain due to upsampling operations in GAN architectures, providing a new perspective for automated identification of fake images.	Images	CelebA and LSUN	[145]	2020	Artifacts observed in GAN architectures in this domain pose challenges that need to be addressed for more robust image analysis.
LSTM and CNN	Works on the colors for deepfake detection.	Videos	FF++, Celeb-DF, and DFDC	[158]	2020	Cannot be applied to the web based application.
Encoder-decoder	Finds multiple artifacts for temporal regions.	Videos	UADFV, FF++, and Celeb-DF	[159]	2021	No effective performance for the compression.
Multi-modal Multi-scale Transformer (M2TR)	An M2TR is utilized for deepfake forensics, which finds local discrepancies at various sizes.	Images/ videos	FF++	[160]	2022	No generalization is achieved.
FrePGAN	The deepfake detector performs better overall when frequency-level variations are detected at the very start of iterations and image-level inconsistencies are taken into account later on.	Image/ Videos	ProGAN, FFHQ, LSUN, CelebA, StyleGAN, StyleGAN2, Big-GAN, and CycleGAN,	[146]	2022	Overfitting issues originating from frequency-level artifacts in generated images.
GAN	Enhances the deepfake detection by utilizing frequency level perturbations.	Videos	ProGAN, FFHQ, LSUN, CelebA, StyleGAN, StyleGAN2, Big-GAN, and CycleGAN	[161]	2022	Overfitting issues in deepfake detection are identified to originate from frequency-level artifacts.
Xception	Comprehensive exploration of features in both the frequency and RGB domains, enhancing the detection of forged face images	Image/ Videos	FF, Celeb-DF, and UADFV	[147]	2023	To handle a larger variety of deepfake manipulation techniques are not covered in the current evaluation.
Space-Frequency Interactive based on CNN	Improves the internal feature representation through space-frequency interaction.	Videos	DFDC, FF++ Low Quality (LQ), HFF, and FF++ HQ	[149]	2023	Limitations in capturing alternation traces in low-quality data, resulting in lower Area Under the Curve (AUC).

TABLE 5. (Continued.) Summary of Deepfake detection methods based on frequency.

CNNFast Fourier Transform (FFT)	Understand features in both the amplitude and phase spectra of images.	Images	ProGAN and LSUN	[98]	2024	Overfit to the artifacts present in the training data, leading to sub-optimal performance on unseen sources.
CNN and Frequency-Aware	Attempts to acquire features that are not connected to particular sources, which is necessary for efficient detection in a variety of GAN models	Videos	Celeba-HQ	[162]	2024	Often fail to generalize well across different sources and GAN architectures.

a: BACKGROUND INCONSISTENCIES BASED DETECTION

By examining the background components of a manipulated video or image, deepfake detection utilized background consistency checks for irregularities that might point to tampering. Although deepfake techniques are improving at creating realistic backgrounds and facial features, they frequently have trouble consistently handling background aspects. Background analysis is a helpful technique for detecting deepfakes because of this flaw, which can show up as blurring, unnatural warping, inconsistent illumination, or uneven depth. The background consistencies are based on the Lighting and Shadow Discrepancies, temporal consistency across frames, and depth and focus mismatch.

In order to enhance face forgery detection, the study [175] presented a unique framework known as Critical Forgery Mining (CFM). Through prior knowledge-agnostic data augmentation, CFM uses a fine-grained triplet technique to suppress certain forging trails, enabling a more reliable detection process. It uses a fine-grained relation learning prototype that employs instance and local similarity-aware losses to mine essential information in forgeries. The research article [175] noted that although compression can remove important forging information, it is still challenging to derive discriminative features from heavily compressed data. As evidenced by the baseline model’s inaccurate identifications, the CFM approach may still have limitations regarding generalization to unseen data despite its increased performance. Examining image metadata, which is normally unaffected, is part of the model [176] to help identify modifications. After that, the image is converted to an error-level format. The output layer is based on two neurons, for real and forgery images, which are utilized to evaluate the image’s authenticity using the neural networks and the metadata analyzer’s outputs. The study’s dataset is initially inappropriate for training the model and needs to be modified to detect pixel-level anomalies brought on by counterfeiting. The research highlights the necessity of a single test dataset and continuous performance benchmarking to enhance forgery detection.

3) BIOLOGICAL/PHYSIOLOGICAL SIGNAL BASED DETECTION

Biological/Physiological signal extraction is initially used in medicine to determine whether a patient’s heart rate or

other signals are normal. This allowed the physician to quickly identify any abnormal biological signals from the patient. In the domain of deepfake identification, researchers used it to measure the inconsistencies such as biological level (heartbeat) or physiological level (head position, eyes blinking, or surrounding boundaries) between fake and real video or images based on physiological or biological signals.

Heart activity is reflected in head movements, which can be utilized to determine heart rate, as demonstrated by Balakrishnan et al. [177], which was a prediction based on video streams. A chrominance approach that Tulyakov et al. devised can significantly increase the precision of heart rate estimates [178]. By analyzing the head and face movements, Agarwal et al. [179] identified the existence and strength of particular action units and used SVM to distinguish between actual and fake videos. CNN and RNN were utilized by Li et al. [180] to identify unusual eyes blinking patterns in fake videos.

A method to recognize deepfake videos using mouth movements was presented by Elhassan et al. [181]. The study created a deepfake identification model with mouth characteristics Deepfake Detection using Mouth Movement and Transfer (DFT-MF) and employed a CNN. Two distinct datasets, the VID-TIMIT and forensics dataset, which comprise real and fake videos, are used to identify deepfakes. Face landmarks are detected using a Dlib classifier. With the Dlib library, one may determine a person’s nose, eyebrows, and other facial traits. Frames that show a closed mouth are detected by measuring the distance between lips. The proposed model states the length of a sentence depends upon the number of spoken words. The study also examined speech rate, revealing that people speak between 120 to 150 words per minute. The proposed approach uses speech rate and facial emotions to determine deepfake videos. Jung, T. [182] introduced a novel technology called “DeepVision” that uses eye blink patterns to distinguish between authentic and fraudulent videos. For the investigations, this research created a dataset using patterns of eye blinking. The duration and frequency of an eye blink’s attributes were counted to distinguish between an authentic and fraudulent video.

Cited studies [183], [184] used a “FakeCatcher” algorithm to identify deepfakes in videos. These studies suggested using the FakeCatcher technology to identify synthesized information from portrait videos. This approach relies on the biological signals gathered from facial regions that

are not well retrained in synthetic material over time or space. Various techniques are suggested to enhance the quality of the generated PhotoPlethysmoGraphy (PPG) signal and the dependability of the collecting procedure. Some recommended improvements include chromaticity attributes [185], green channel components [186], optical attributes [187], and Kalman detectors [188].

a: MOTION DYNAMICS BASED DETECTION

Motion dynamics based deepfake detection takes advantage of the natural, coherent motion patterns found in real videos, which are challenging for deepfake techniques to capture throughout numerous frames accurately. The motion dynamic based detection depend upon the both audio/visual and biological/psychological inconsistencies in image or videos. Many deepfake algorithms produce frames one at a time without accurately simulating real human motion's dynamics over time, leading to motion-based inconsistencies. These anomalies can show up as differences in lip synchronization, abnormal blinking patterns, or abnormal head motions, which are helpful indicators for identifying transformed facts. Deepfake detection is based on motion dynamics involving the multi-modal fusion, gaze direction, and lip syncing and speech patterns based. **Multi-Modal Fusion Based Detection** integrates information from multiple sources, such as audiovisual and textual or physiological signals, to reveal the consistencies.

A novel deepfake detection method utilizing Multi Modal Fusion (MMF) and Dense Convolutional Neural Networks (DenseCNN)s was proposed by the paper [189]. The method overcomes the shortcomings of current systems by improving sensitivity to complicated manipulations and optimizing feature reuse through dense connectivity patterns. InstructBLIP, a vision language model, is incorporated into the process of identifying deepfake multimedia by offering thorough explanations for discovered modifications. A classification layer that detects the existence of deepfake content comes after multimodal separation and is used to extract several modalities from videos. DenseCNN is then used to extract features. The model's capacity to generalize to other kinds of deepfake alterations that aren't represented in the training data may be limited by its dependence on particular datasets. Audio Visual Dual Transformers based in Dynamic Weight Fusion is introduced in article [190]. The model amplified both intra and cross-modal forgery cues to improve detection capabilities. It uses a face transformer and an audio transformer encoder in a two-stage process to record the spatial features of facial emotions. The model overcomes the difficulties presented by multi-modal deepfake detection by combining multi-modal conversion with dynamic weight fusion to accommodate heterogeneous input from audio and visual modalities. According to the paper, the performance of baseline approaches has been affected by the minimal amount of videos in the DF-TIMIT dataset. **Gaze Direction Based Detection** takes advantage of the eye movement patterns

and natural alignment found in real videos, which makes it difficult for deepfake techniques to recreate accurately. In real videos, the gaze direction exhibits subtle, organic movements corresponding to the head posture, facial alignment, and surrounding context. Gaze direction also considers the eye blinking, eye movement, and gaze head pose consistency. One of the major limitations is that the gaze direction-based deepfake detection models require high-quality data.

An eye-tracking database named FakeET is used in the study [191] to examine how people perceive deepfake videos visually. The 811 videos in the collection, which was assembled from the Google deepfake dataset, are 331 real and 474 fake. Forty users viewed each video, with each film being viewed at least twice. The videos are divided into segments to examine eye movements and gaze patterns and capture temporal consistencies. The study is based on user-centric evaluations, which is not a proficient approach for demographics and circumstances complexities, as is the case for deepfake detection. Cartella et al. [192] proposed a statistical analysis-based technique to detect the fake image. Cartella et al. highlighted various types of manipulated images in terms of realism, contributing that instruction-guided alterations are more accessible for people to spot than semantic-aware modifications. The method is affected by the features of the images, such as poor quality or objects that are not in context, which impacts the realism evaluations.

Lip Syncing and Speech Patterns Based Detection In order to identify any tampering, deepfake detection evaluates the lip movements that correspond with audio data and speech patterns' sustainable flow. It is challenging to achieve realistic lip sync in deepfakes, so the variations in speech and lip movement or an irregular speech flow are proof of manipulation. These techniques are very effective in detecting of face-swapped images or audio-deepfakes.

The research study [193] uses a face-swapping technique, which is based on a GAN, to create Deepfake videos. For every 32 participants, two models were trained: a HQ model with a 128×128 size and a LQ model with a 64×64 input/output size. Korshunov et al. used publically accessible pre-trained VGG and facenet architectures for face recognition, using the bottleneck and facenet-7 layers as characteristics for validation. The face recognition confidence score is determined using the cosine distance between feature vectors. As part of the evaluation methodology, the equal error rate, a widely used system performance indicator, was calculated by dividing the false acceptance rate by the false reject rate. The robustness of the results may have been impacted by the lack of a development set for hyperparameter adjustment in the VidTIMIT dataset used for testing and training. The approach [194] combined a boundary-based image color hashing technique with the Minimum Covariance Determinant (MCD) approximation. The goal of this combination is to identify minute irregularities in deepfake videos. Four separate stages compose the methodology, guaranteeing a methodical workflow for video frame analysis. Although the technique is excellent at identifying abnormalities, the

research did not detail the particular datasets that were utilized, which could restrict how broadly the results can be used. The author Kharel et al. [195] utilized visual cues from the video, improving detection accuracy by synchronizing lip movements with the input audio through a cross-attention method. The model focused on pertinent facial movements by using a transformer encoder for face self-attention and a fine-tuned VGG-16 network for visual feature extraction. The method used MLP head that utilize the outputs from the audio and video transformer encoder modules to produce the final classification results. The Ahmed et al. [196] suggested that MLP is utilized to classifying image received by the transfer learning model. The class imbalance in deepfake datasets is a major drawback that can make it more difficult to develop an objective model. According to the authors Kharel et al. [195] future research must use bigger and more balanced datasets or strategies to address class imbalance.

b: HEART RATE VARIABILITY (HRV) BASED DETECTION

Heart Rate Variability (HRV) based deepfake detection uses the microscopic changes in skin tone and face texture to assess physiological signals like pulse rates that are challenging for deepfake algorithms to replicate accurately. Even a slight blood flow variation that correlates with heartbeats is the source of the mild color changes observed on human faces in the original video. In contrast, deepfakes lack a natural pulse-based pattern and usually focus on realistic facial characteristics rather than physiological information. The approach based on the remote photoplethysmography (rPPG) to detect fake videos is proposed by Boccignone et al. [197], exploring the complex structure of heart rate dynamics retrieved from face video streams. The approach depends on the agent-in-context (AIC) theory, arguing that physiological signals are not manipulated in fake videos compared to real nature. The approach estimated the Blood Volume Pulse (BVP) signals from face patches and predicted artificial manipulation using two sets of characteristics. The complexity is measured by intra patch BVP and designed to capture physiological behavior. Relying only on physiological signals leads to a loss of transparency, which is not a practical approach for deepfake detection. Moreover, it is still liable after additional interferences, such as head movements or changes in lighting. The research study [198] employed a Beats Per Minute (BPM)s analysis approach based on rPPG for deepfake detection. BPMs are computed from videos to quantify the fluctuations in blood volume based on skin color. The variations in BPMs between real and fake videos revealed deepfake. The study highlighted that deepfake detection models based on rPPG still have detection accuracy and generalization performance limitations.

c: HEAD POSE INCONSISTENCIES BASED DETECTION

The goal of head pose inconsistencies analysis for deepfake detection is to spot irregular or strange head movements that happen when faces are artificially edited. The orientation of a

person's head in three dimensions, as defined by angles like pitch (up-down), yaw (left-right), and roll (tilt), is referred to as head pose. Deepfake models find it challenging to reliably recreate the head motions that naturally connect with eye direction, body posture, and surrounding context in real-world recordings. The Compact Ensemble-based Discriminator (CED), which detects spectral anomalies during the GAN up-sampling procedures, is trained using a technique [199] that divides real-time video frames into segments. CED provide a strong training signal by enabling it to discriminate between real and fake images. Because many deep generative models, including DCGANs, require large amounts of data to operate efficiently, which might not always be available, the dependence on high-quality training data may be a drawback. In order to improve the identification of deepfake videos, the author Lai et al. [200] suggested a multi-feature fusion detection approach that extracts features from many domains and directly extracts features from face images using the Xception network. Utilized the Xception network to interpret facial images after applying the discrete DFT to produce spectrum maps. The Xception network is utilized to extract features from images. LSTM network for temporal analysis used the features from the first three domains to classify the deepfake. The input videos' quality and features could substantially impact the model's performance, suggesting that it has to be further improved under various circumstances.

D. SPATIO-TEMPORAL DOMAIN BASED DETECTION

Some researchers focused on the spatial and temporal domains or relied on one of them, spatio-temporal, as shown in Table 7. Both are more influenced by detecting deepfakes because they observe minor manipulation in realistic images or videos. A 3DCNN classifier that may acquire properties of the spatio-temporal criteria from the 3D images was suggested by Nguyen et al. [221]. They employed 3D convolution kernels to build a deep 3DCNN that learned spatio-temporal properties from short subsequent frame series to identify the deepfake. In their technology, the characteristic mappings in the CNN layers are detached from the frames of the preceding layer. Thereby, it is able to collect facial data in several frames in both spatio-temporal dimensions. The proposed scheme was performed on VidTIMIT and Faceforensics datasets up to 99%. Bansal et al. [222] proposed a Recycle-GAN technique based on cycle-GAN; spatial and temporal data were merged with the support of GAN. Due to this, the merging of spatial and temporal data led to an effectual output. The computer tomography module aimed to snippet characteristics at a higher level with less input. To capture the up-sampling artifact of face forgeries, spatial phase shallow [223] learning integrates the phase spectrum in DFT with the spatial image.

3DCNN were used as a spatio-temporal module to combine spatial data into the temporal domain. To increase the spatio-temporal module's performance, Steganalysis Rich

TABLE 6. Summary of Deepfake detection methods based on temporal features.

Classifier/ Technique	Key Features	Dealing	Dataset	Ref	Year	Limitations
Long-term Recurrent Convolutional Network (LRCN)	Detects blinking pattern frequency in deepfake.	Videos	49 videos and corresponding deepfakes, CEW	[181]	2018	There is no performance for synthesized fake videos.
Gaussian Mixture Model (GMM), SVM, MLP, and LSTM	Exploring various methods for lip-syncing and dubbing detection	Videos	VidTIMIT, AMI, and GRID	[172]	2018	Computational complexity associated with processing large amounts of video and audio data for feature extraction.
LSTM	CNN is used for frame level feature extraction and then sends them to LSTM for conducting the sequences frame for classification.	Videos	HOHA and 300 random videos	[165]	2018	Need to improve its robustness against unseen manipulated videos.
Capsule networks	Detects forgery videos and images. Detects replay attacks. Extraction from the VGG-19 network is used in the capsule network for grouping.	Images/ Videos	FaceForensics and replay attacks record	[202]	2019	There is no performance for noise detection. Low performance for mixed attacks.
SVM	Works on facial features and head pose to detect the deepfake	Videos	Popular personalities videos	[180]	2019	Has no performance for the live interviews.
CNNs	Explores the differences between the naturally generated video frames.	Videos	FF++	[203]	2019	Works for limited data.
SVM	Manipulated speed of video could be helpful for deepfake detection.	Videos	2019 Deepfake Identification Challenges	[168]	2020	Not effective for existing manipulated video frames.
SVM	Based on the biological artifacts and is used to discriminate between realistic and fake videos.	Videos	FF, FF++, Celeb-DF, and UADFV	[204]	2020	No effectiveness for complex videos.
GAN DeepVision	Examining notable variations in the blinking pattern, which is a spontaneous, voluntary motion that does not involve cognitive effort.	Videos	Kaggle	[183]	2020	Rely on only specific media for feature extraction.
XceptionNet	Identifies deepfakes that involve subtle modifications, such as lip-syncing and, audio and mouth synthesis and replacement based on partial words.	Videos	A2V, T2V-S, and T2V-L	[173]	2020	Needs for further refinement to encompass all visemes in the analysis, which could enhance detection outcomes even more.
CNNs	Based on audio related to video inconsistencies and modalities find the deepfake	Videos	DFDC and DF-TIMIT	[170]	2020	Focuses on only a specific person.
RNN	Detection of manipulated frames with an automatic weighting mechanism.	Videos	DFDC	[205]	2020	Getting better performance, audio analysis is also included with this method.
CNNs	Works on pixel intensity and gradient pixel information.	Videos	FF++, DeeperForensics (DFo)1.0, and Celeb-DF	[206]	2021	Has no training for unseen manipulated data.
LSTM, RF, XG-Boost, Logistic Regression (LR), and KNN	Emotions are used for deepfake detection, expressed by the face and face.	Videos	SEMAINE and DFDC	[207]	2021	Only applicable to British regions.
LSTM	Deepfake detection based on gazes and eyes.	Videos	FF ++, Celeb-DF, and DFo++	[208]	2021	LSTM is not effective for smaller datasets.
CNNs	Check the two eyes' similarities.	Videos	FFHQ and StyleGAN2 Images	[209]	2021	Working only on pixel differences could not contribute to the geometry and scene.
LR	Lips movement concerning the jaws, mouth, and pairs of ears.	Videos	YouTube downloaded videos	[210]	2021	Stuck on ruffled hair. Head unwavering movement and closed eyes.
CNNs	Finds the dissimilarities between the inter frames' structure.	Videos	FF++	[211]	2021	Utilizes only one dataset .
RNNs	Temporal detection is made on the geometric features such as facial land-marking.	Videos	UADFV, FF++, Celeb-DF, and DFo1.0	[212]	2021	Unable to make effective prediction for longer geometric temporal.
CNNs	Appropriate noise features and semantic discrepancies.	Videos	FF++	[213]	2021	Not effective for unseen manipulated data.

TABLE 6. (Continued.) Summary of Deepfake detection methods based on temporal features.

Robustness Analysis Technology Forecasting (RATF)	Automatically generates corresponding temporal filters, allowing for the comprehensive capture of various inconsistencies in deepfake videos	Videos	FF++, DFDC, Celeb-DF, and WildDeepfake	[164]	2022	Intra snippet information ignored. Traditional temporal convolution. Limited temporal modeling.
LSTM	Focuses on local low-level cues and temporal information to enhance generalizability across deepfake methods.	Videos	CelebDF, DFDC, FaceSh, DeepFo, and FF++	[214]	2022	Manipulations and its performance in real-world scenarios is not explicitly discussed in the provided contexts.
Cascaded Deep Sparse Autoencoder	Focuses on reducing the overfitting problem in feature selection.	Videos	F2F and DFDC	[215]	2022	Utilizes high resources for computation.
Hierarchical Contrastive Inconsistency Learning (HCIL)	To improve inconsistency learning, the framework includes two modules: one for intra-snippet inconsistency mining (region-adaptive) and the other for cross-snippet information fusion (inter-snippet fusion).	Videos	FF++, Celeb-DF, DFDC, and WildDeepfake	[216]	2022	The HCIL framework, while outperforming existing works on datasets like DFDC, still faces challenges in dealing with the varied duration of videos in the Wild-DF dataset, impacting overall performance.
GAN	Geometry is used in face mesh and face segmentation to preserve the face shape. Based on the lip-syncing audio input with face attribute so that alteration could easily be detected.	Audio/ videos	Voxceleb	[217]	2023	Only for limited users.
RNN	Encodes identity information in all frames of a video clip to identify vectors and learns temporal identity embeddings to determine video authenticity	Videos	FF++	[218]	2023	Due to the nature of close-set baselines, making a fair comparison with TI 2 Net is difficult.
ResNet50 and LSTM	Determines the authenticity by precisely identifying distinct faces in the video.	Videos	DFDC, Face Forensics, and Celeb-DF	[219]	2023	Requires high computational resources for training and running the DNNs.
DNN and ResNet50	Analyzes discrepancies in the surfaces of the captured scene over time induced by deepfake manipulation.	Videos	Real and fake Videos	[220]	2024	EfficientNet-b0 performance implications are suggested for future research.
Xception, StyleConvolutional Gated Recurrent Unit (ConvGRU), PatchForensics, and Face X-ray	Detecting fake videos by analyzing style latent vectors and their temporal changes in generated videos.	Videos	FF++, Celeb-DF, FSh, and DFo	[221]	2024	Xception, have limitations in capturing temporal cues present in deepfake videos, potentially affecting detection accuracy.

Model (SRM) features were gathered from the input frames and used to hide the frame content while enhancing its appearance. In addition, the computer tomography module used the few variables feasible for obtaining deep-level characteristics from the spatio-temporal module's output.

The convergence and generalization capabilities of 3DCNN were negatively impacted by their higher parameters compared to typical 2D CNN. In order to reduce the number of variables, the Computed Tomography (CT) module was designed to take the role of the conventional 3D convolution layers. The outcomes of their study showed that their network behaves better on large deepfake datasets than modern deepfake detection methods while having fewer variables than other networks.

1) GENERAL DNN BASED DETECTION

Recently popular mechanisms for deepfake generation include deep learning techniques like GANs and autoencoders, frequently utilized in the synthetic image

domain. These techniques are utilized to examine a person's facial expression and actions to replicate another person's facial emotions and movements using AI, so more robust techniques are required for these synthesized media. Hence, previous research does not count the forensics model generation; it only focuses on the trained datasets and their models. To overcome this issue, the forensics CNN has been proposed by Sun et al. [224]. In order to identify forged human photos, this approach uses Gaussian Noise and Gaussian Blur in two phases. According to this technique, high-frequency pixel noise in low-level pixel statistics can be improved, while modest high-frequency information artifacts in deepfake photos can be neglected. By adopting this behavior, GANs images can easily be classified, so differentiation between real and fake images can be quickly made.

Sabir et al. [225] suggested a scheme utilizing the RCNN. There are two steps in this scheme. Face alignment and cropping were extracted using a spatial-transformer network

called the face manipulation stage. The RCNN was used for face manipulation detection based on prior output in step two. Step two also discriminated against information recorded from the manipulated frames of videos. A spatio-temporal encoder investigates the inconsistencies within visual and audio elements at both temporal and spatial levels. Compared to this, an end-to-end technique named Audio Video (AV)FakeNet was presented by Ilyas et al. [226] by taking into account both the audio and visual features of a video.

a: DUAL ATTENTION BASED DETECTION

Deepfake detection algorithms effectively evaluate visual artifacts and their temporal evolution with the help of a dual attention mechanism, a complex approach that integrates spatial and temporal attention. Dual attention enhanced the model's ability to identify minor discrepancies that could indicate image and video alteration by focusing on both domains.

The study [227] introduced a Dual Attention Forgery Detection Network (DAFDN) to enhance the identification of faked videos. The Spatial Reduction Attention Block (SRAB) focused on key spatial features in video frames. The Forgery Feature Attention Module (FFAM) module found certain clues of video manipulation. Integrating these attention processes enables the convolutional neural network to effectively eliminate inconsistent traces from images, which is crucial for detecting deepfakes. The main feature extractor in the model is EfficientNetV2 (ENV2), which first utilizes its pre-trained weights and maintains its layers frozen to benefit from its broad feature extraction capabilities. A Convolutional Block Attention Module (CBAM) used both Spatial Attention (SA) and Channel Attention (CA) methods to improve feature representation, is incorporated into the method. CA customized feature responses at the channel level, whereas SA focused on specific regions within feature maps. Fine-tuning, which required unfreezing deeper layers of ENV2 to customize learned features, maximizes performance against deepfake datasets. The proposed network is validated by precise experiments on four benchmark datasets: DFDC, FF++, Celeb-DF, and World Leaders. The efficacy of networks against developing synthetic media manipulation techniques may be impacted by the dependence on intermediary feature extractors without enough empirical validation and the absence of generalized optimization strategies. The article [228] suggested an Xception model that incorporates feature fusion and a dual attention mechanism, especially for face forgery detection. Three primary flows make up the architecture are as entry, middle, and exit. While the middle flow employs the CBAM for feature refining and is improved to capture high semantic features through different convolution levels, the Entry flow stays the same from the original Xception model. The Exit flow improves the model's classification capability by learning global and local information from the fused features using a self-attention

mechanism and depth-wise separable convolution. Although the suggested approach performs better, the paper does not cover certain drawbacks or difficulties encountered during implementation or assessment.

V. DEEPAKE DETECTION DATASETS

Datasets provide the basis for training, testing, and benchmarking in the context of deepfake detection. Developing and accessing deepfake algorithms heavily relies on the availability of authentic, diverse datasets. In addition, it is easier for detection techniques to reach their maximum detection capabilities and accurately represent their methods' performance when more datasets are added. A collection of some significant datasets is listed in Figure 9.

A. UADFV

Yang et al. [110], [180] produced a dataset for their deepfake detection tests. They used the FakeApp program to create 49 phony videos after gathering 49 actual videos from YouTube. In the authentic footage, they replaced Nicolas Cage, an American actor, with a different visage. As a result, he is the sole persona that appears in every phony.

B. DeepfakeTIMIT

Derived from the VidTIMIT dataset [259], DeepfakeTIMIT is a video dataset [21]. For face swapping, the researchers used the open-source FaceSwap-GAN technique. A total of 16 couples were manually chosen, and 620 face swap-based deepfake videos from thirty-two people were assembled despite altering the initial audio channel of the recordings. The two main quality standards for fake videos are LQ photos with 64×64 pixels and HQ photos with 128×128 pixels.

C. WildDeepfake

A dataset called WildDeepfake [260] is used to detect deepfakes in real-world scenarios. It is made up of 7,314 face frames that have been taken from 707 deepfake films that have been gathered entirely from the internet. A tiny dataset called WildDeepfake may enhance existing datasets and create more potent detectors against real-world deepfakes.

D. FaceForensics

With almost 500,000 face-containing frames generated from 1004 clips, FaceForensics [261] is a video dataset that may be utilized for investigating photo and video forgeries. Every video is obtained from YouTube and edited into brief, continuous snippets, most featuring frontal faces. There are two distinct versions of this dataset.

1) SOURCE-TO-TARGET

In this technique, the developer reenacts over a thousand videos using fresh facial expressions taken from other videos. This may be used, for example, to train a classifier to identify fake photos or videos.

TABLE 7. Summary of Deepfake detection methods based on spatio-temporal features.

Classifier/ Technique	Key Features	Dealing	Dataset	Ref	Year	Limitations
Recycle-GANs	Creates input/output pairs using adversarial learning's cycle-consistency criterion to train supervision in tasks like image-to-image translation.	Videos	Viper dataset	[223]	2018	Lack of correspondences between training samples from the input and output domains.
Gated Recurrent Unit (GRU) and CNN	Temporal inconsistencies with the facial features and Landmarks.	Videos	FF++	[226]	2019	Only works for FF++ compression version.
AlxNet	Deepfake detection using head pose, a siamese based network classifier.	Videos	VoxCeleb2	[230]	2019	Does not work on complex datasets with triplet loss.
CNN	Actions depend upon the facial expression and head movement that could be learned from facial and behavioral from ResNet101 VGG, respectively.	Videos	FF++, Google/Jigsaw, DFDC, and Celeb-DF	[231]	2020	No work for complex and unseen data.
LSTM and CNN	Work on heartbeat for image verification.	Video	DFDC-preview and FF++	[232]	2020	Do not work on unseen data.
CNN	Lip-sync indicates deepfake.	Videos	MOBIO and GRID	[233]	2020	Visualize authentication leads this technique toward poor performance.
Attention Map network	Frames levels are used as deepfake detection.	Videos	Videos from Youtube and Celeb-DF	[234]	2020	Audio modality could be fake, which is not considered.
LSTM and CNNs	Linguistic features of frames uses deepfake detection.	Videos	Celeb-DF	[235]	2020	Drawback of this mechanism is to convert frame level models into video level models by passing frame level results through a median filter with a window size of five to reduce false positives.
Transformer-encoder	Identification of landmarks over an extensive range.	Videos	DFDC, Celeb-DF, and FaceForensics	[236]	2021	Has no focus on the unseen images manipulated.
Autoencoder and GAN	Focused on local texture information over high-level semantic information, reducing receptive fields to suppress high-level features and concentrate on the local region.	Videos	FF++, Celeb-DF, and DFDC	[224]	2021	Overfitting, limiting their effectiveness to specific training datasets and facing challenges in real-world scenarios.
SVM	Minute facet depiction and comparatively simple classifiers are used to identify the manipulated video sequences.	Videos	FF++	[237]	2021	No effective for blurry videos.
3DCNN and SVM	State of art methods in detecting manipulated facial videos created by deepfake technologies.	Videos	FF++ and VidTIMIT	[222]	2021	Lack of attention to Spatio-temporal features often results in lower accuracy of the model.
CNNs	By using the attention mechanism, temporal features are detected.	Videos	Celeb-DF and FF++	[238]	2021	Not effective for compressed videos.
LSTM	Temporal discrepancies between the frames.	Videos	FF++	[239]	2021	Works only on high quality videos.
CNNs	Detects the frame level inconsistencies.	Videos	Celeb-DF and FF++	[240]	2021	More robustness is required for compressed videos.
CNNs	The prototype network helped to detect the temporal discrepancies in frames.	Videos	FF++, DFo, and Celeb-DF	[241]	2021	Some artifacts, such as subtle discolorations or movements, are challenging to interpret with just one image, highlighting a limitation in the interpretability of certain visual cues.
CNN based on encoder	By adopting the Latent Pattern Sensing (LPS), a predictive representative learning technique is used to identify deepfake videos.	Videos	DFDC and Celeb-DF(v2)	[242]	2022	Overfitting, especially in the DFDC dataset is highlighted as a potential limitation.
CNN and LSTM	Effectively learned rich spatial features and temporal inconsistencies, showcasing competitive results even with a reduced frame number.	Videos	DFDC, FF++, and Celeb-DF	[243]	2022	Focus on a limited number of frames, miss the vital information.

TABLE 7. (Continued.) Summary of Deepfake detection methods based on spatio-temporal features.

3DCNN	Frame classification loss functions to detect temporal forgeries effectively.	Videos	Localized Audio Visual DeepFake (LAV-DF)	[244]	2022	Audio reenactment method used in the dataset does not always generate the reference style.
RNN	Focus on improving deepfake video detection through the utilization of metric learning to enhance feature extraction quality.	Videos	FaceForensic++	[245]	2022	Resource limitations influence the choice of applying contrastive loss and training the network on hard examples.
XceptionNet and 3DCNN	Extracts independent local spatial and temporal features using a two-stream CNN and then fully connected layers process the extracted features for classification.	Videos	CelebDF and FF++	[246]	2022	The ability to identify extremely complex deepFake movies that closely resemble actual faces could still have certain limits.
CNN	A spatio-temporal Enhancement block to learn regularity disruptions across space and time, enhancing the detection of deepfakes.	Videos	FF++, DeepFake Detection Challenge (DFDCP), Deepwild, and Celeb-DF	[247]	2022	The performance of models may drop when testing on deepfakes generated by unseen techniques.
ConvGRU, 3DCNN, and RNN	A temporality-level stream is utilized to obtain temporal correlation characteristics, and a frame level stream is utilized to learn spatial inconsistency from phase based reconstructed frames.	Videos	FF++ and DF-1.0	[248]	2022	Do not use the frequency domain information or other representation spaces of video frames to their full potential.
ResNet	The approach integrates texture enhanced shallow features with deeper features in the spatial stream and reinforces frame level feature learning through a spatial attention mechanism.	Videos	FF++ and DFDC	[249]	2022	Neglected to pay attention on small, well-localized pattern modifications in both space and time that are distinctive to a certain operation.
CNN	Detects inconsistencies in mouth movements to enhance robustness against adversarial attacks and perturbations.	Videos	DFDC	[250]	2023	Attention mechanisms used to guide networks in focusing on inconsistencies are still vulnerable to various image impairment processes, voluntary perturbations, and adversarial attacks, which are increasingly common.
F3Net	Detecting facial manipulations across various datasets.	Videos	FF++ LQ F2F, FF++ LQ NT, DFDC, FF++ LQ DF, Celeb-DF, and FF++ LQ FS	[251]	2023	Overfitting for known manipulation types.
Self-supervised Transformer	Using a spatio-temporal generator, the model is learned to overcome domain shift problems and acquired generalizable representations in an unsupervised manner.	Videos	FSH, DFDCP, and DFDC	[252]	2023	The limited generalization power of supervised deepfake detection approaches is shown in their inability to consistently maintain good performance in datasets including undetected modification techniques.
Multi-Scale Transformer framework	Considers temporal cues often overlooked by frame based methods, improving the overall detection capabilities.	Videos	DFDC, Celeb-DF-v2, DF1.0-Raw, and FF++	[253]	2023	Could investigate the generalization of the MSVT framework to even more diverse forgery techniques and datasets to assess its robustness in detecting emerging deepFake methods.
AVFakeNet framework	Identifies multimodality using an innovative method.	Video	Celeb-DF, Fake Audio Video Celebrities (FakeAVCeleb), LA, World Leaders dataset, Automatic Speaker Verification (ASV)Spoof-2019, and Presidential deepfakes dataset	[227]	2023	Requires highly computational resources.
Spatiotemporal Trident Network (STN)	Utilizes the CBAM to enhance the network's focus on meaningful video information, improving detection accuracy.	Images	UADFV, Celeb-DF, and FF++	[254]	2023	There is a research gap as video based detectors have not been studied as much as image based detectors.
DNNs	Proposed two methods, Directed Acyclic Graph (DAG) and True Positive Rate (TPR) and fairness metrics may be improved by optimizing hyperparameters with certain indices, demonstrating the efficacy of the suggested techniques in enhancing detection accuracy and fairness.	Videos	FF++, DFDC, and Celeb-DF	[255]	2024	The research's suggested methods are predicated on the ideas that each occurrence of a loss function is independent and that loss functions may be broken down into separate terms.

TABLE 7. (Continued.) Summary of Deepfake detection methods based on spatio-temporal features.

VIT (Encoder)	The temporal module uses a latent presentation layer and the patch-temporal multi-head self-attention. The spatial module uses the Facial Component Guidance (FCG) loss to create an accurate depiction from each frame's important face features.	Videos	FF++, CelebDF-V2, DFDC, , and DFo	[256]	2024	Generalization is not achieved.
Encoder	Utilizes the temporal facial features and Temporal Coherence Analyzing Network (TCAN) and has two parts: first is for head feature vision and second for a transformer encoder for feature extraction and result prediction.	Videos	FF++, DeepFakeDetection, Deepfake Detection Challenge, Celeb-DF-V2, and DFo1.0	[257]	2024	Examines several color spaces to improve the temporal modeling design and include audio modalities for improvement.
CNN with VIT	Both local and global information leads to superior performance on unseen datasets	Videos	FF++	[258]	2024	Suffers from poor generalization under new deepfake generation methods due to their focus on local spatial artifacts.
CNN	Temporal coherent positional embedding for temporal information encoding and size embedding for capturing relative face sizes, leading to improved deepfake detection performance on the forgeryNet dataset.	Videos	ForgeryNet	[259]	2024	One limitation of the forgeryNet dataset is the absence of labeled test data, which necessitated using the validation set as the test set for evaluation.
MesoNet, FWA, Xception, and capsule network	Incorporate of Texture Analysis Model (TAM) and Attribute Analysis Model (AAM) to enhance the representation of spatio-temporal feature differences between real and forged faces.	Videos	FF++	[260]	2024	Overfitting issues leading to poor cross dataset detection performance.

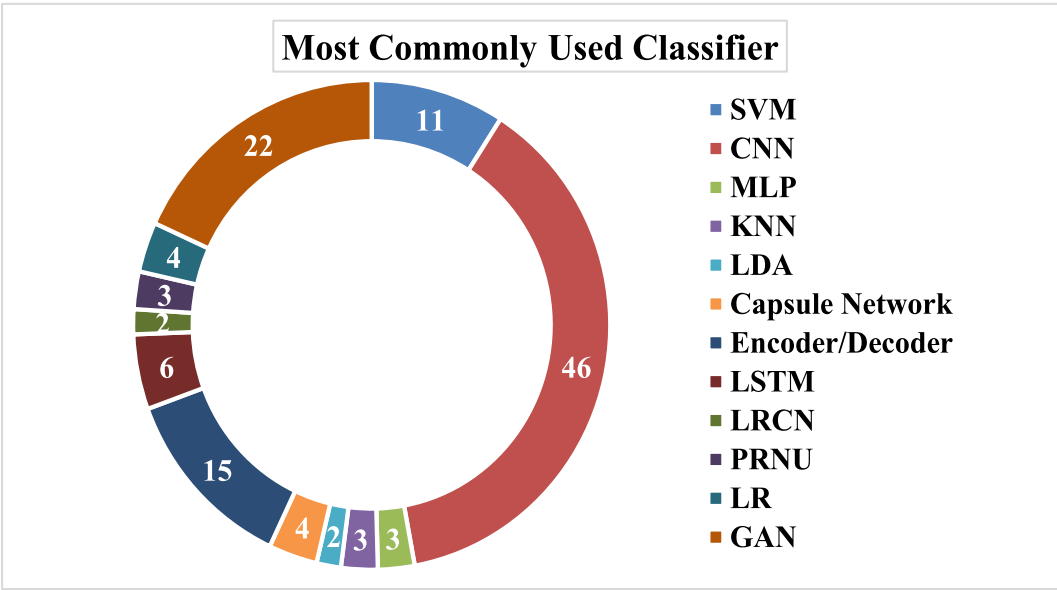


FIGURE 8. Most commonly used classifiers extracted from literature.

2) SELF-REENACTMENT

In this technique, F2F is utilized by researchers to mimic the facial expressions in movies using their own expressions as input. The result is a pair of videos that may be used, for example, to train supervised generative refinement models.

E. FaceForensics++

FF++ [108] is a type of forensics dataset composed of over 1000 real video clips that have been altered using the F2F, deepfakes, NeuralTextures, and FaceSwap automated face modification methodology. The data was derived from 977 YouTube videos, all of which have recognizable,

mostly frontal facial features free of obstacles, which made it possible to create realistic-looking forgeries using automated manipulating technologies. Rossler et al. [108] provided binary masks; the data might be used for segmentation as well as image and video classification. Furthermore, they offer 1000 deepfakes algorithms to produce and enhance fresh data.

F. CELEBRITY-DF

The Celeb-DF dataset [262] consists of 795 synthetic videos produced by deepFake and 408 genuine movies. The videos in the Celebrity-DF dataset all have the usual 30 FPS frame rate and a typical duration of 13 seconds. The genuine videos are gathered from YouTube and divided into two groups. There are 158 videos in the initial collection featuring 13 celebrities of various genders and skin tones. There are 250 movies in the second series, each covering a distinct topic. Real1 and Real2, respectively, are the names of two sets. To decrease or eliminate visual artifacts seen in prior deepFake datasets, an improved version of the deepFake generation algorithm was used to create our own collection of synthesized films from the genuine video.

G. DFDC

With over 100,000 video clips, the DFDC [263] is a deepfake detection dataset. There are two versions of the DFDC dataset that are used for facial manipulation: 1) a preview dataset containing 5k videos and algorithms and 2) a complete dataset having 124k videos. Eight facial manipulation algorithms are used. Gender, skin tone, illumination, head positions, age, and backdrop are numerous in both versions.

H. FakeAVCeleb

FakeAVCeleb [264] is a new audio and video deepfake dataset that includes corresponding synthesis replicated audio with additional deepfake videos.

I. DeeperForensics-1.0

With 60,000 movies, including an aggregate of 17.6 million frames, DFo-1.0 [265] is undoubtedly the largest dataset offered by facial fake detection, ten times larger than similar datasets. 11,000 modified videos and 48,475 source videos make up the whole dataset. The modified videos are produced using a recently suggested many-to-many face swapping or end-to-end technology called (Densely Fused-Vocational Auto Encoder) DF-VAE. The real videos were collected from consenting performers from 26 countries, and 100 were paid. Seven different real-world perturbation types at five different intensity levels guarantee a more significant size and greater diversity.

J. KoDF

The Korean Deepfake (KoDF) detection Dataset [266] is a comprehensive database of artificially generated & real videos highlighting the Korean subjects utilized for deepfake

identification processes. The collection includes 175,776 fake videos from 403 participants and 62,166 genuine recordings.

K. WaveFake

A dataset for detecting audio deepfakes is called WaveFake [267]. The collection is made up of more than 100,000 produced audio snippets.

L. ForgeryNet

He et al. [268] created the massively expensive ForgeryNet dataset, which includes unified labels in video and images like information from four operations: 1) method of classifying images forgeries is the two-way method (real/fake), the three-way method (fake with identity-remained forgery methods, fake/real with identity replaced forgery methods), and the n-way method (actual fifteen corresponding techniques for forgeries), 2) Localization of Spatial Forgeries: this technique divides the fabricated picture's altered region in relation to the matching genuine image source, 3) Video Forgery Classification: This technique uses randomly positioned, modified frames to redefine the video-level forgery classification. This work is crucial since attackers may change any target frame in the real world, and 4) Temporal Localization of Forgeries, which helps to pinpoint the modified temporal spans. In the sense of data scale (221,247 videos, 2.9 million images), modifications (7 image and 8 video level strategies), disruption (36 distinct & more combined disruption), and annotations (2.9 million modified area annotations, 221,247 temporal forgery segment labels, and 6.3 million categorization labels). ForgeryNet is the largest deepfake forgery dataset that is publicly accessible.

M. DFDM

Jia et al. [269] generated 6,450 deepfake videos using various Autoencoder models, which we collected into a new dataset called Deepfake Videos Generated from Different Models (DFDM). Five Autoencoder models were used to produce deepfakes using the same input (encoder, decoder, intermediate layer, and input resolution).

N. DeePhy

Three distinct generation algorithms were used to create the 5040 deepfake videos that comprise the innovative deepfake phylogeny dataset, DeePhy [270]. This is among the initial datasets based on Deepfake Phylogeny, illustrating the steps of producing deepfakes through the sequential implementation of different generation algorithms. The dataset may be utilized for the following tasks: (i) DeepFake Detection, (ii) Deepfake Model Attribution, and (iii) Detection of the deepfake methods' sequential sequence used to generate phylogenetic deepfakes.

O. DeepFake MNIST+

A deepfake face animation dataset is called DeepFake Modified National Institute of Standards and Technology

(MNIST)+ [271]. An image animation creator for State Of The Art (SOTA) creates the dataset. 10,000 distinct facial animation clips are included, and they may be used to confuse the latest livens detectors.

P. AUDIO VISUAL-Deepfake1M

Extremely pragmatic deepfake audio visual information is difficult to detect and locate, even with the most sophisticated advanced methods. The majority of research endeavors in this field concentrate on identifying superior deepfake images and videos; nevertheless, a limited number of studies tackle the issue of localizing short fragments of audio visual modifications into authentic videos. Cai et al. [272] presented the AV-Deepfake1M datasets. This dataset covers content-driven (i) video modifications, (ii) audio modifications, and (iii) audio visual modifications resulting from more than 1M videos.

Q. DEEP-VOICE

Examples of realistic human speech are included in the [273] DEEP-VOICE; Deepfake Voice Recognition (Jordan Bird) dataset, along with Deepfake variations of those speeches produced by retrieval-based speech conversion. Retrieval-based Voice Conversion was used to transform the speech of eight well-known people into one cohesive voice for DEEP-VOICE. The background noise was removed using the Retrieval based Voice Conversion (RVC) for each speech after adding the original accompaniment to the deepfake speech.

VI. POTENTIAL CONCERNS AND FUTURE DIRECTIONS

After carefully reviewing the existing literature, we observed that the future of deepfake technology is poised to be shaped by both advancements and challenges. The impacts of deepfake were discussed in Figure 5. It is predicted that this dynamic technology will be widely accepted in many sectors, such as education, entertainment, medicine, and trades and emphasizes that handling future deepfake technology will be very difficult due to the rapid development of AI, deep learning, and machine learning. However, the potentials concerns and direction of deepfake in the future are highlighted below:

A. POTENTIAL CONCERNS

The following are the potential concerns of the deepfake.

- 1) In the future, deepfake could allow the manipulation of the stock market and would benefit the malign actor to generate an illicit profit [83].
- 2) Deepfake might be utilized to generate fake images of natural disasters (earthquakes or flood areas), and a large amount of funds could be obtained [274].
- 3) Mimicking appearances, expressions, and voices could be a crucial step in developing virtual assistants who look like real humans and whom we can communicate with. With the help of digital influencers, called virtual

humans, they are gradually but steadily making their way into the mainstream and could influence how we engage with one another. Though storytellers produce their content, digital influencers don't reply to you in their own words, but they are a sign of things to come in the future in which interactions with genuine virtual entities will be more "natural." Thus, deepfake technology, which is trained on an enormous amount of human behavior data, may enable intelligent assistants to comprehend and generate highly sophisticated dialogue [16].

B. FUTURE DIRECTIONS

Following are the potential future directions.

1) FUTURE OF DEEPPAKE IN PREDICTION

- 1) In the future, criminal investigation could be possible by deepfake using the strategic deployment of synthetic media in terms of the methodical collection of data in considering ethical issues [275].
- 2) Deepfake could produce fake weather forecasting and climate-changing reports that might ruin celebrities' or politicians' plans and disrupt national policies and procedures [276].
- 3) Due to the malicious usage of deepfake, there is a key benefit: it could improve governance and authentication mechanisms [277].

2) FUTURE OF DEEPPAKE IN MEDICAL TECHNOLOGIES

- 1) In the future, the deepfake could help the blinds to track their way by using the camera that takes live photos of the track and then recognizes these images with the help of trained deepfake detection models; after that, the distance will be calculated, and information will transfer to the blind's ears, in this way the low vision could navigate [278].
- 2) Deepfake could be used in healthcare for the diagnosis of disease and to keep track of medical records of patients [279].
- 3) Students of medical universities can diagnose and treat simulated patients, whereas commerce students can replicate critical transactions without the risk of consequences in the real world [280].

3) FUTURE OF DEEPPAKE IN GRAPHICS AND ENTERTAINMENT

- 1) Deepfake could be used to validate copyright protection through hashing and digital watermarking [281].
- 2) The future of the deepfake is that it allows the announcements to be made in different languages simultaneously, and the movies will not need subtitles [282].

In the future, deepfake may take over in other fields. Deep learning techniques produce a person's fingerprints more accurately, and deepfakes would be utilized to produce reports and tweets more significantly.

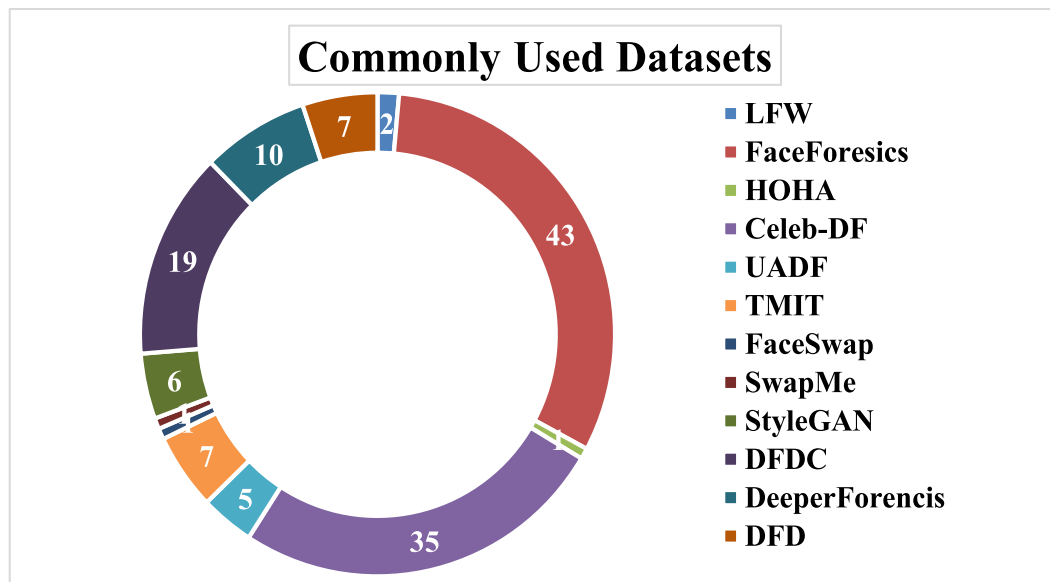


FIGURE 9. Most commonly used dataset extracted from the literature.

VII. CONCLUSION

Throughout this survey paper, we addressed a detailed systematic literature review on deepfake technology for insight view of the technological transformation in deepfake. In this regard, we focused on a comprehensive overview of the deepfake generation process using GANs and encoder-decoder pairs. The classification of deepfake was based on four categories: synthetic face, facial reenactment, facial attribute alterations, and face swap. Current tools for deepfake generation and beneficial and malicious applications of deepfake were also discussed in this review. A detailed empirical study was also performed concerning the deepfake detection classifiers, their key features, and their limitations. The currently available datasets for deepfake detection, most commonly used in deepfake technology, were also described in detail. This article will help owners and developers of deepfake detection techniques to enhance their models by upgrading knowledge and incorporating certain datasets known to yield better results for the particular model type they work with. Well-defined approaches are utilized; however, in the deepfake detection process, heavy augmentation affects the detection accuracy and requires heavy-duty consumption resources for computations, so much attention is needed to generate robust techniques considering data augmentation for deepfake detection. As the deepfake generation process is relatively easy and is generated through mobile apps now a days, there must be attention to developing mobile apps that will be proficient for end users to detect deepfake. Despite substantial improvements, there are a lot of issues with the present detection techniques that must be resolved to detect deepfakes properly. Real-time deepfake detection and utilization of detection techniques in many industries and on various channels would be a significant challenge because

of its complications, including the computational capacity required for deepfake detection in real time, given the tremendous amount of information sent on the internet every second. The requirement is that these detection techniques work well with very few false positives. In order to achieve this goal, one can use sophisticated learning techniques such as meta-learning and metrics, as well as efficient designs, such as compression techniques and transformers, for example, quantitation, and determined investments in a solid foundation of hardware and software. Furthermore, the selection or rejection criteria process may have led to some errors in sorting these studies. In this article, we have gathered the relevant papers from several digital repositories, publications, workshops, and conferences. It is still conceivable that our compilation of research studies does not include all relevant publications.

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